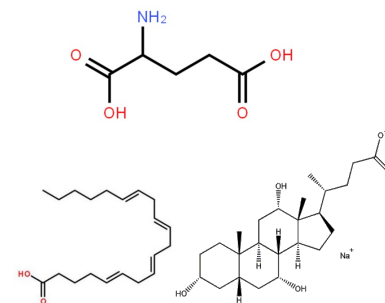
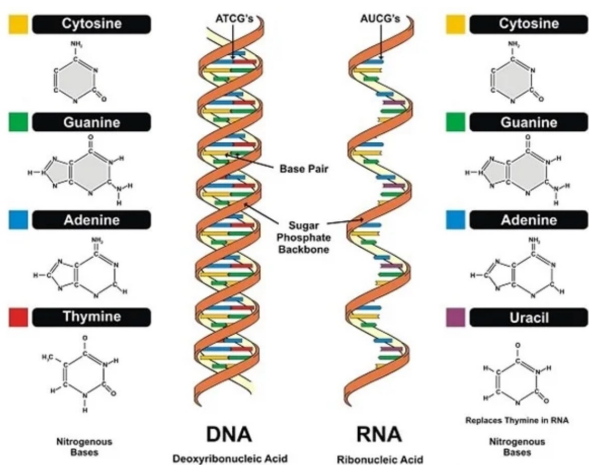
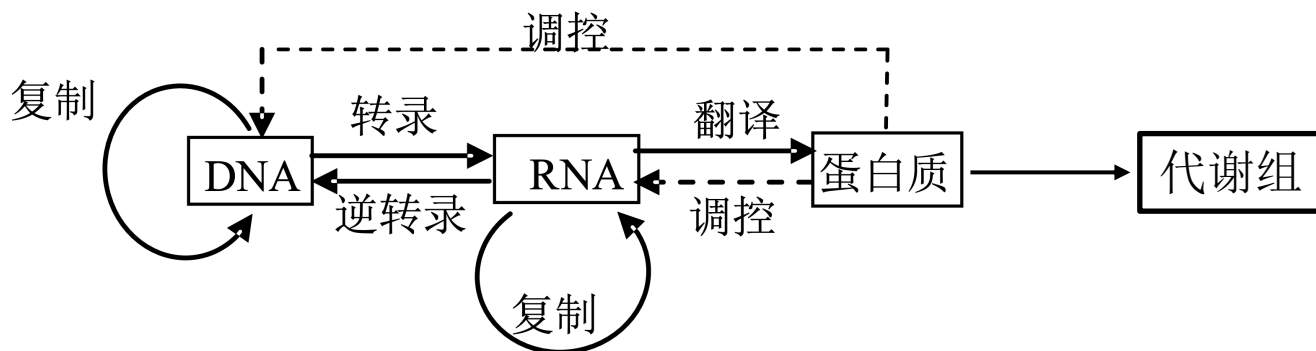

蛋白质组、代谢组技术与数据分析

李雪媛

2026.04.14

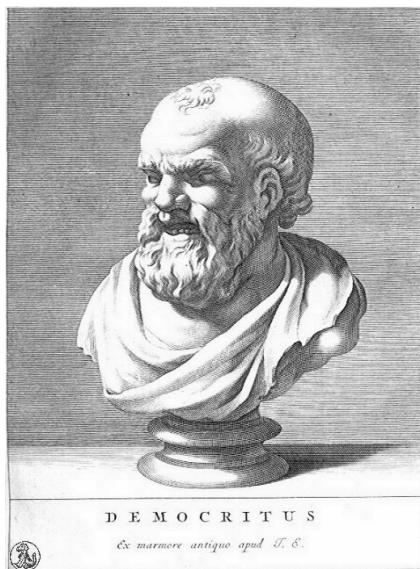
功能基因组与蛋白质组

■ 基因组计划及局限性



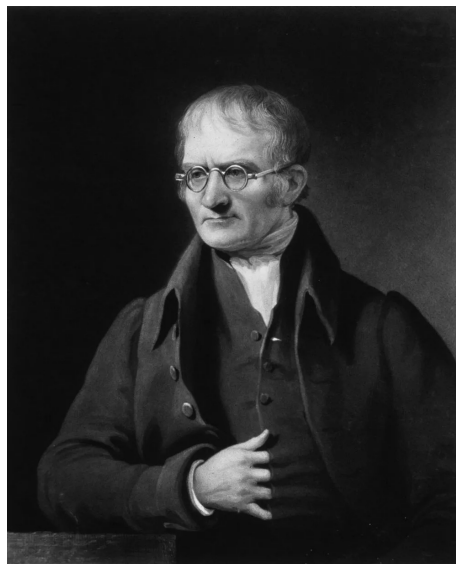
物质的组成

《庄子*天下篇》：“一尺之棰，日取其半，万世不竭。”



德谟克利特 Δημόκριτος
公元前460年～370年

宇宙是由虚空和原子组成
原子是不可再分的微粒



John Dalton
1766年～1844年

现代化学原子论
元素就是原子

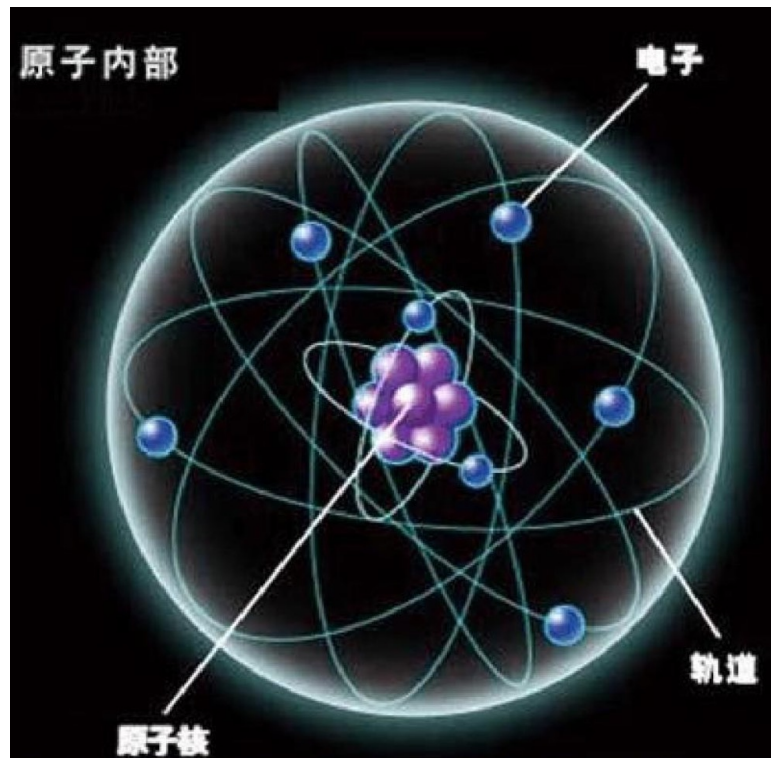
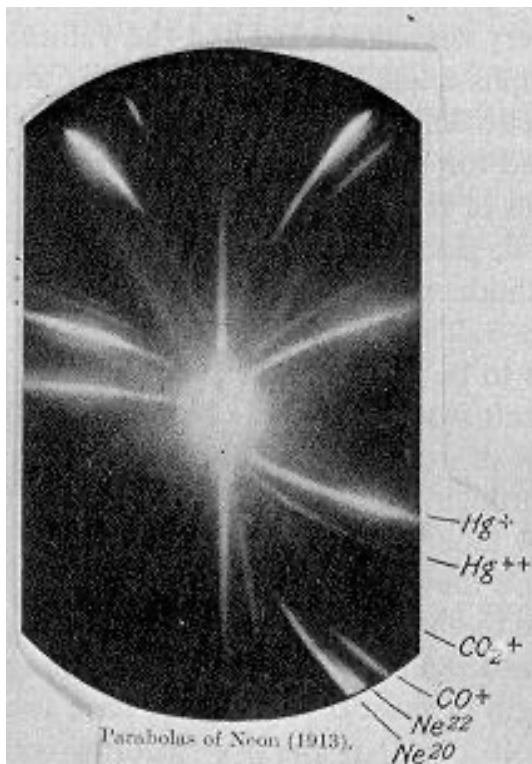


Joseph John Thomson
1856年～1940年

发现电子
证明同位素存在

蛋白质组研究方法概述

Thomson证明同位素存在



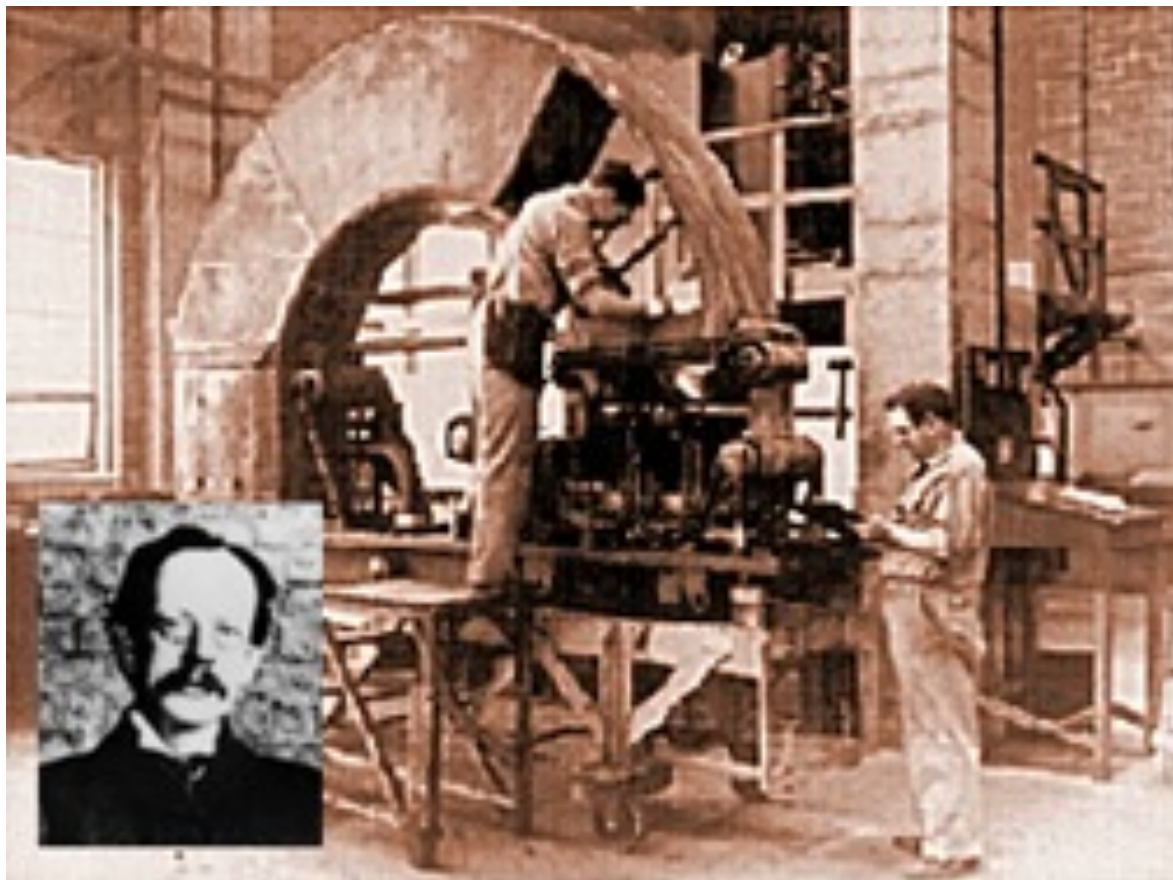
质子 中子

蛋白质组研究方法概述



Francis William Aston
1877年~1945年

发明质谱仪
证明大量同位素存在



质谱仪工作流程

- 电离：将待分析物质中的化合物分子或原子通过化学或物理方法进行电离，使其成为带电离子。
- 加速：利用电场将带电离子加速到高速度，以便能够进入下一个步骤。
- 分离：将加速后的离子通过磁场或电场进行分离，得到不同的离子种类。
- 检测：将分离后的离子进行检测，记录其质量-荷比值 (m/z) 。
- 分析：将记录下来的质谱图与数据库进行比对，确定物质的分子或原子质量以及结构。

现代质谱仪结构

■ 质谱仪组成



液相进样系统

气相进样系统

固相进样系统

EI、CI、ECI、
ESI、MALDI

TOF、Q、
IT、ECR

电子倍增器
光电倍增器
平板式微通道板检测器

■ 质谱仪的主要指标

质量范围--根据检测的对象选择

分辨率--分开相邻两个离子的能力

灵敏度--检测微量成分的能力

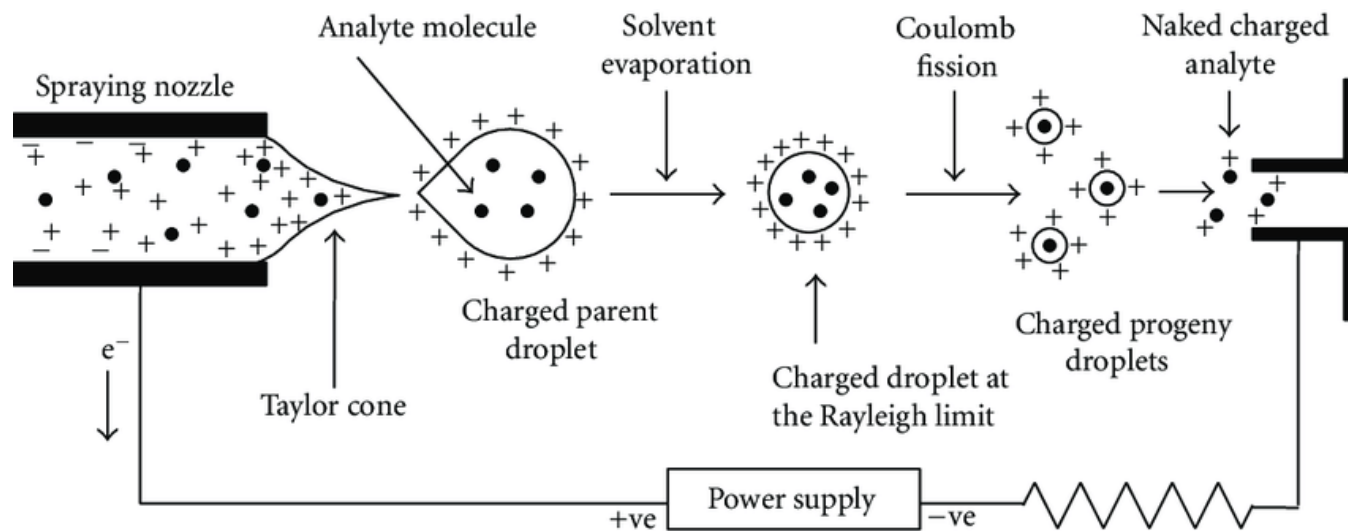
质量精度--测量离子质量的精确程度

现代质谱仪结构—离子源

- **离子源**
- **电子轰击离子源 (Electron Impact, EI)**：是最常用的离子源之一。它使用高能电子（通常为70 eV）将化合物分子电离，并在离子源中形成分子离子。该离子源可以提供相对丰富的离子流量，并且具有较高的离解能，能够产生高能离子碎片，因此通常用于分析**有机化合物**。
- **化学电离离子源 (Chemical Ionization, CI)**：使用化学反应将化合物分子电离。它通常使用较低能量的电子（通常为10 eV）将反应气体（如甲烷、乙烷等）电离，形成离子对，然后这些离子对再与分析物质发生反应形成分子离子。CI离子源对**烷基、芳香族**和其他化合物的分析具有优异的选择性和灵敏度。
- **电喷雾离子源 (Electrospray Ionization, ESI)**：用于分析**高分子量化合物和生物分子**（如蛋白质、多肽和核酸等）。在该离子源中，将待测物质溶解在含有挥发性的有机溶剂中，并通过高电压喷嘴喷出，形成一个带电荷的喷雾。溶剂的挥发性使离子带有电荷的残留物形成分子离子，从而进行分析。
- **电化学离子源 (Electrochemical Ionization, ECI)**：它使用电化学反应产生离子。通常在分析物质溶于电解质中的情况下使用，通过施加电位，在电极上产生离子。
- **金属表面辅助离子源 (Matrix-assisted Laser Desorption/Ionization, MALDI)**：用于分析**大分子量的生物分子**，如蛋白质、多肽和核酸。在MALDI离子源中，分析物质与一种基质混合，形成一个分子和基质的复合物。然后，通过激光的照射，这些分子和基质复合物会被蒸发并离子化，形成分子离子。

现代质谱仪结构—离子源

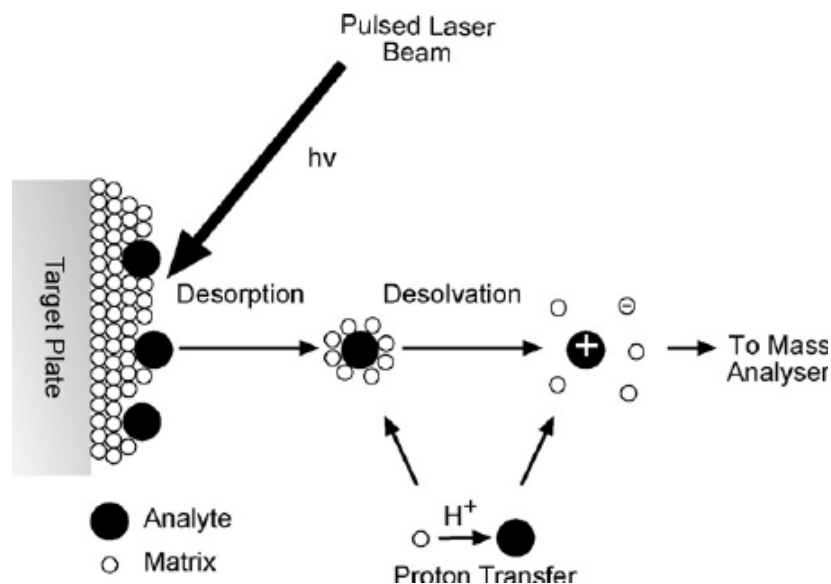
- 电喷雾质谱技术(Electrospray Ionization Mass Spectrometry , ESI - MS)
- ESI适用于小分子和大分子的分析，但对于大分子的分析灵敏度比较低。



现代质谱仪结构—离子源

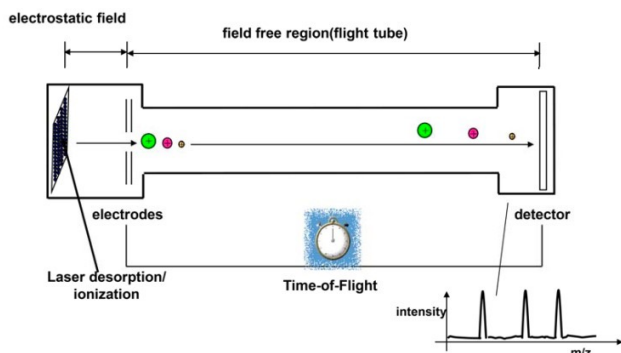
■ MALDI

- 利用激光脉冲将样品分子与基质分子共同离子化，MALDI适用于大分子的分析，具有较高的灵敏度和离子化效率，适用于高通量和成像分析等应用场景

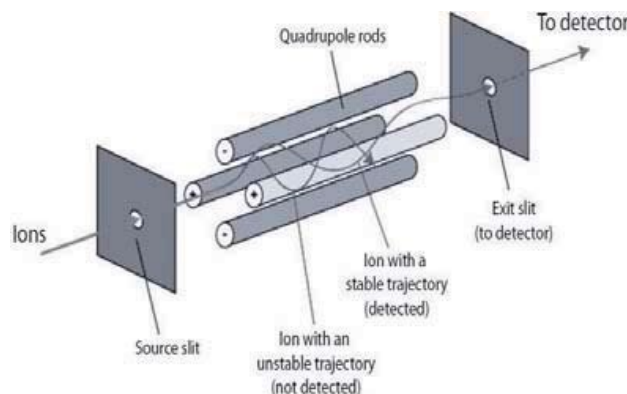


现代质谱仪结构—质量分析器

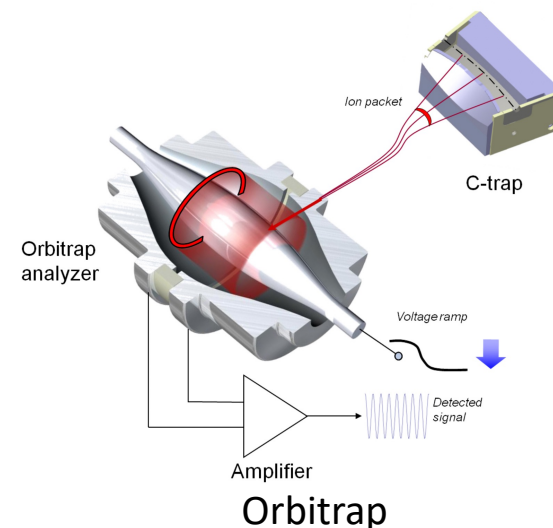
- 时间飞行质量分析器 (Time-of-Flight Mass Analyzer, TOF)
- 四极杆质量分析器 (Quadrupole Mass Analyzer, Q)
- 离子阱质量分析器 (Ion Orbitrap Mass Analyzer, Orbitrap)



Time-of-Flight Mass Analyzer, TOF



Quadrupole Mass Analyzer, Q



Orbitrap

MALDI-TOF质谱仪



蛋白质组研究方法概述

■ 蛋白质组的研究历史

Marc Wilkins于1994 提出 “Proteome”

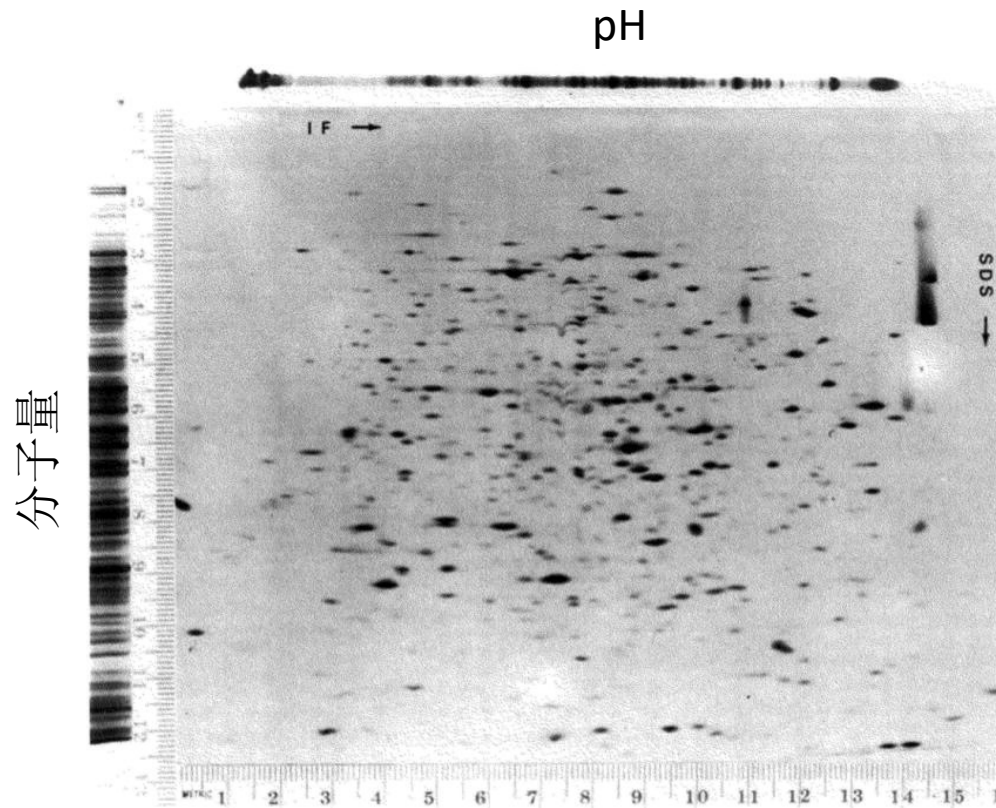
1995年，悉尼大学Humphery Smith实验室与Williams等4家实验室合作，在文献中首次公开使用 “proteome”一词

■ 蛋白质组的含义

一个基因组、一种生物或一种细胞/组织所表达的全套蛋白质

蛋白质组研究方法概述--2-DE

■ 二维凝胶电泳(two—dimensional gel electrophoresis, 2—DE)



蛋白质组研究方法概述--2-DE

■ 二维凝胶电泳图像处理

常用软件：PDQuest 6.0、Image Master 2D Elite、Phoretix 2D

- 1)蛋白质点检测(spot detection)
- 2)背景消减(background subtraction)
- 3)归一化处理(normalization)
- 4)蛋白质点匹配(spot matching)

蛋白质组研究方法概述--2-DE

■ 二维凝胶电泳的局限性

1) 实验设计

10^6 - 10^7 个细胞，样本间异质性，细胞异质性

2) 重复性

蛋白提取、预处理、条件优化、仪器稳定性

3) 动态范围

细胞中蛋白拷贝数 10^1 - 10^8

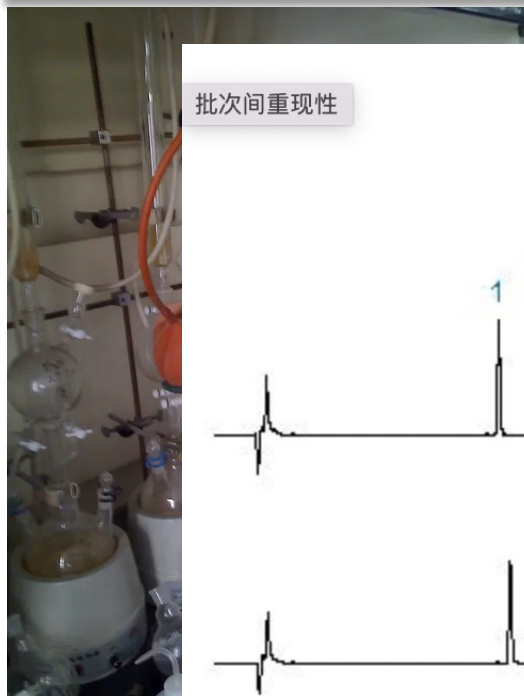
4) 蛋白质丢失

疏水性蛋白高/低分子量的蛋白

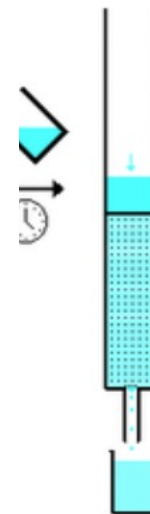
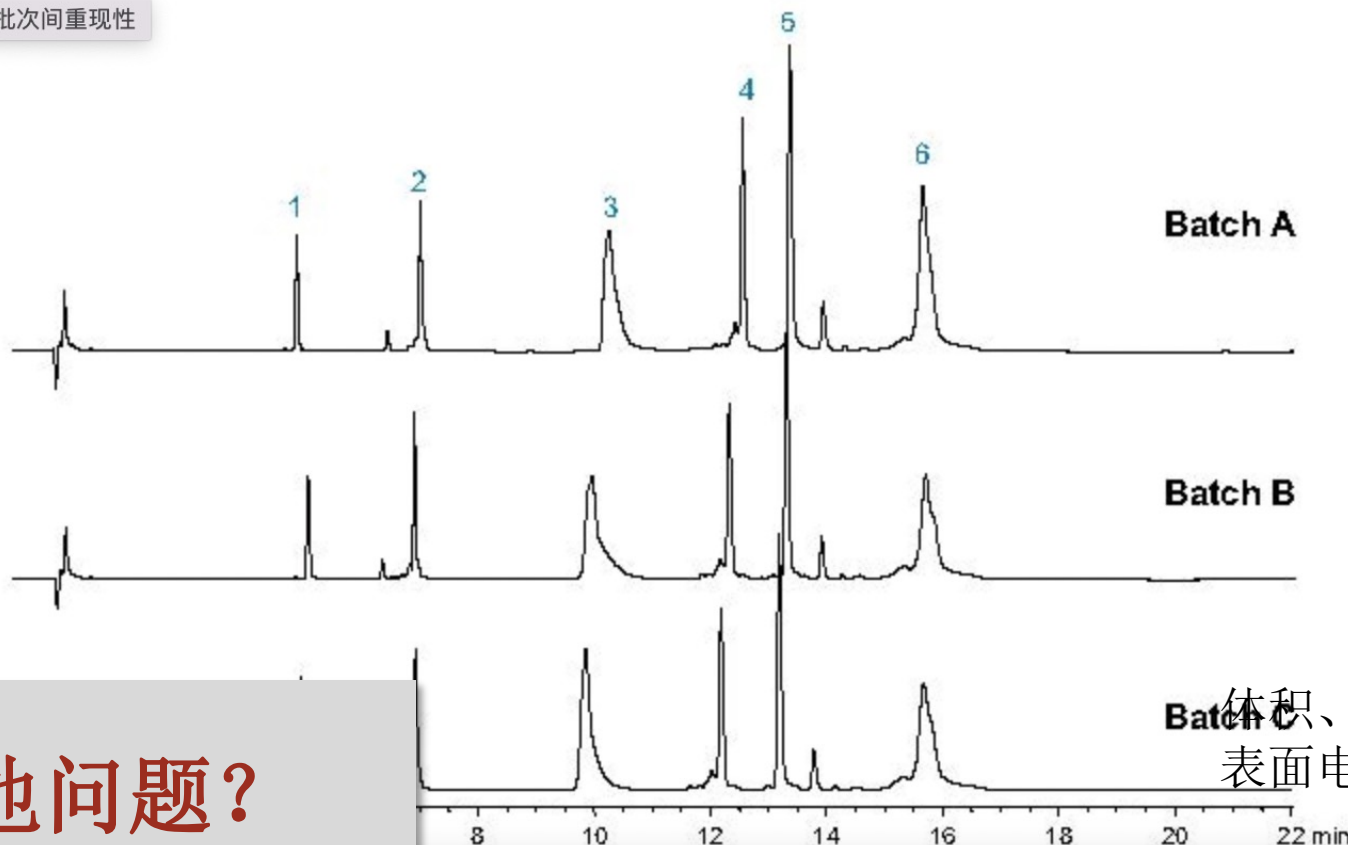
5) 通量和自动化

实验流程，图像分析处理

蛋白质组研究方法概述--色谱



批次间重现性

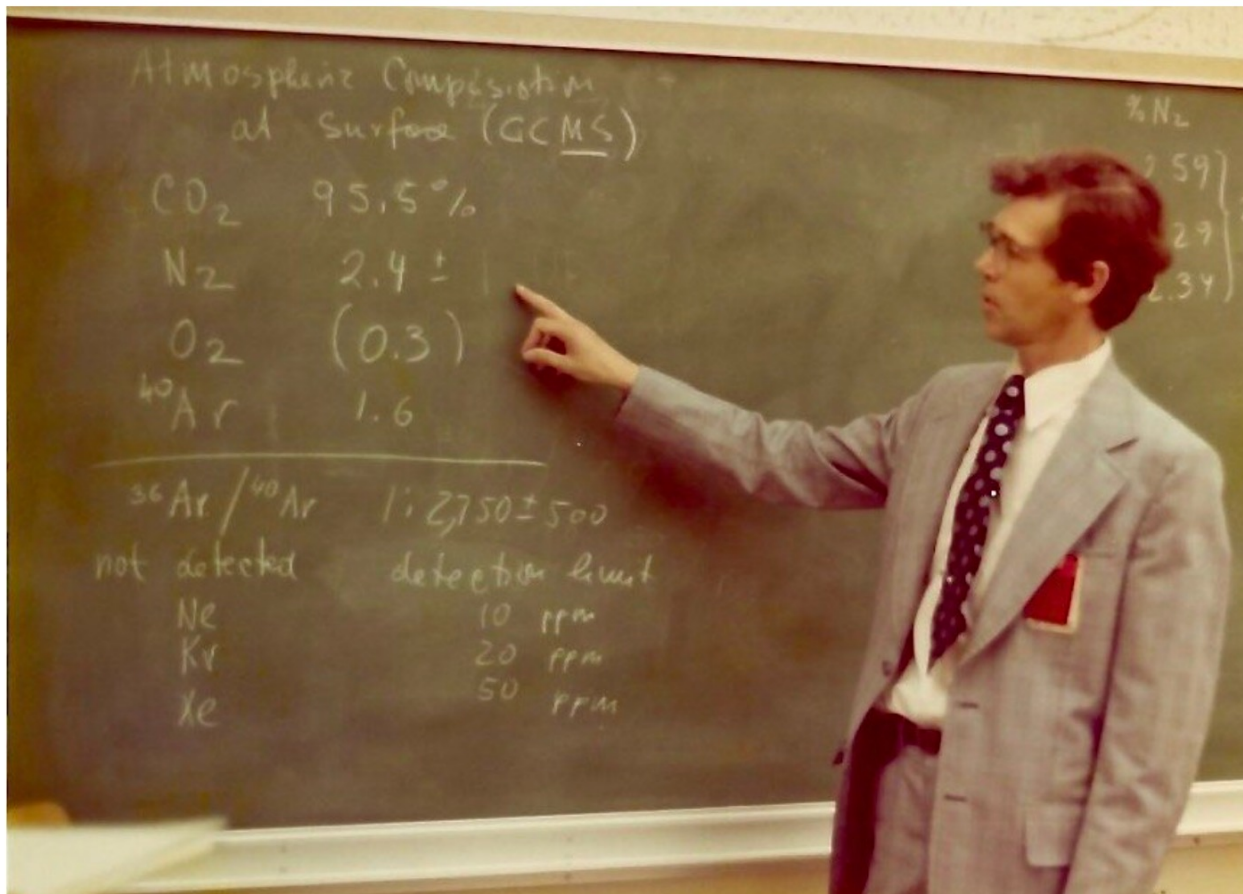


体积、亲疏水性
表面电荷

其他问题？



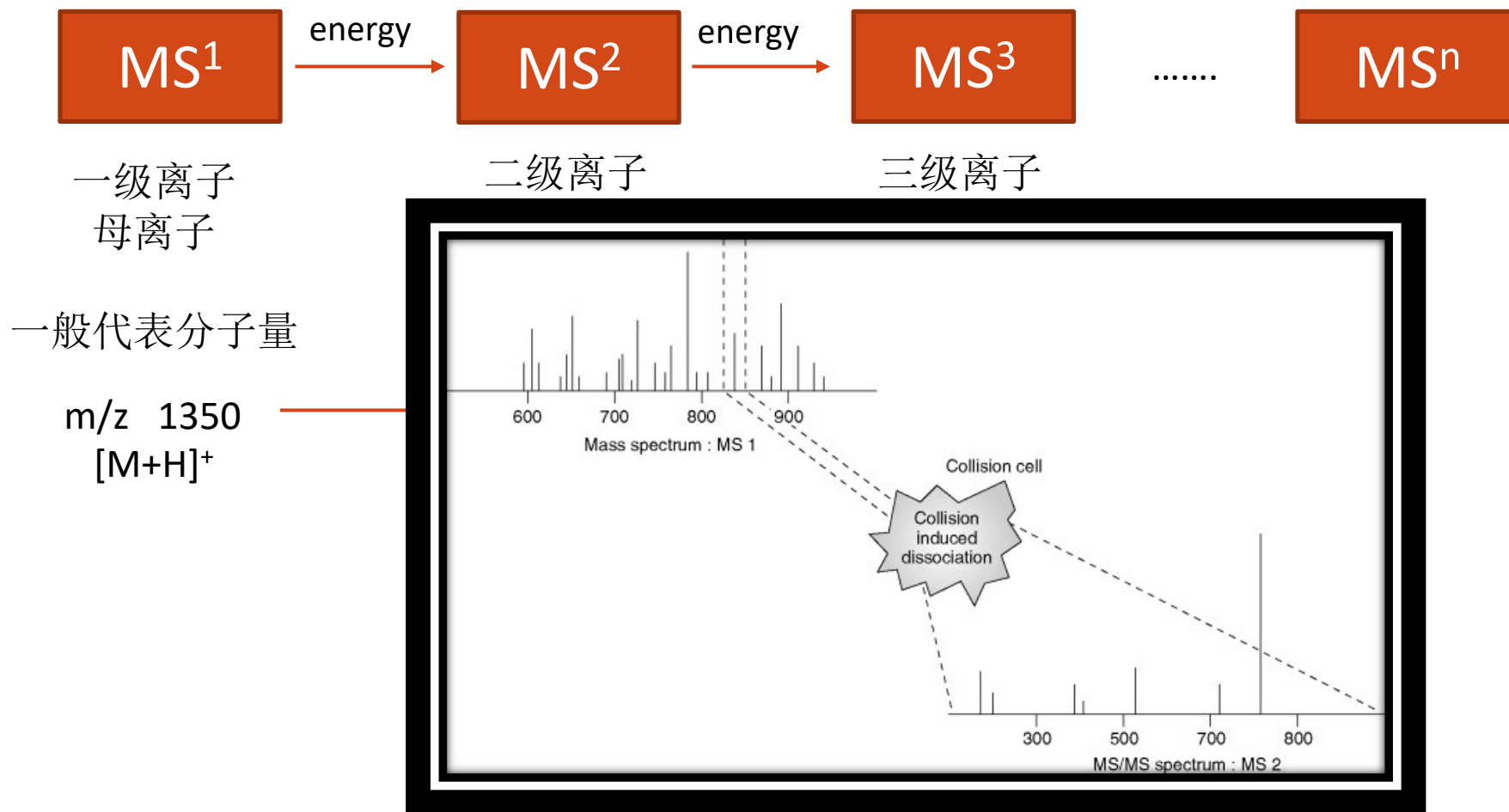
有机质谱



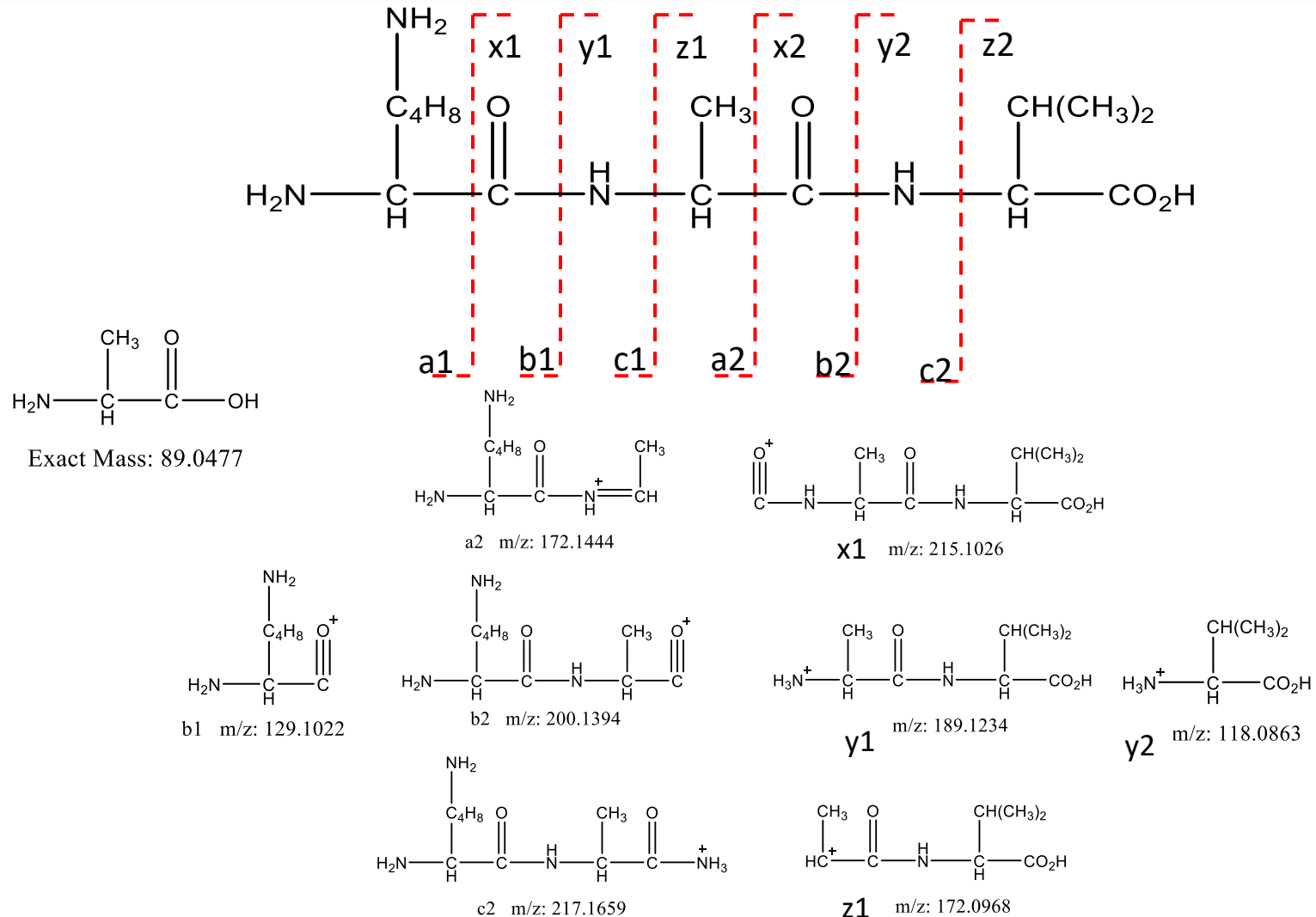
Klaus Biemann 1926年~2016年

有机质谱之父

串联质谱

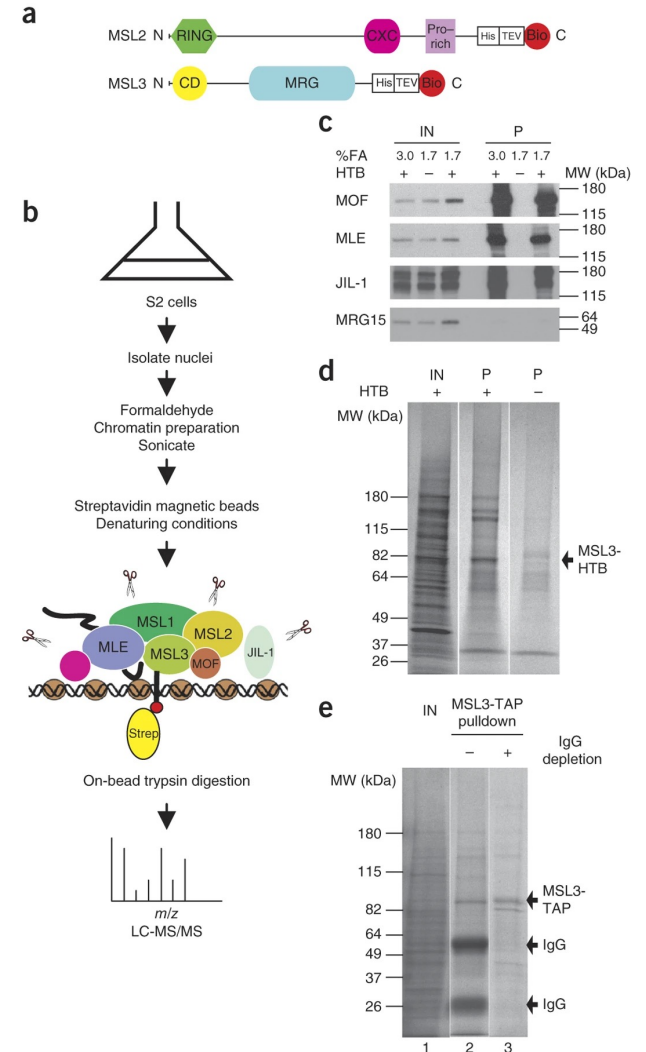
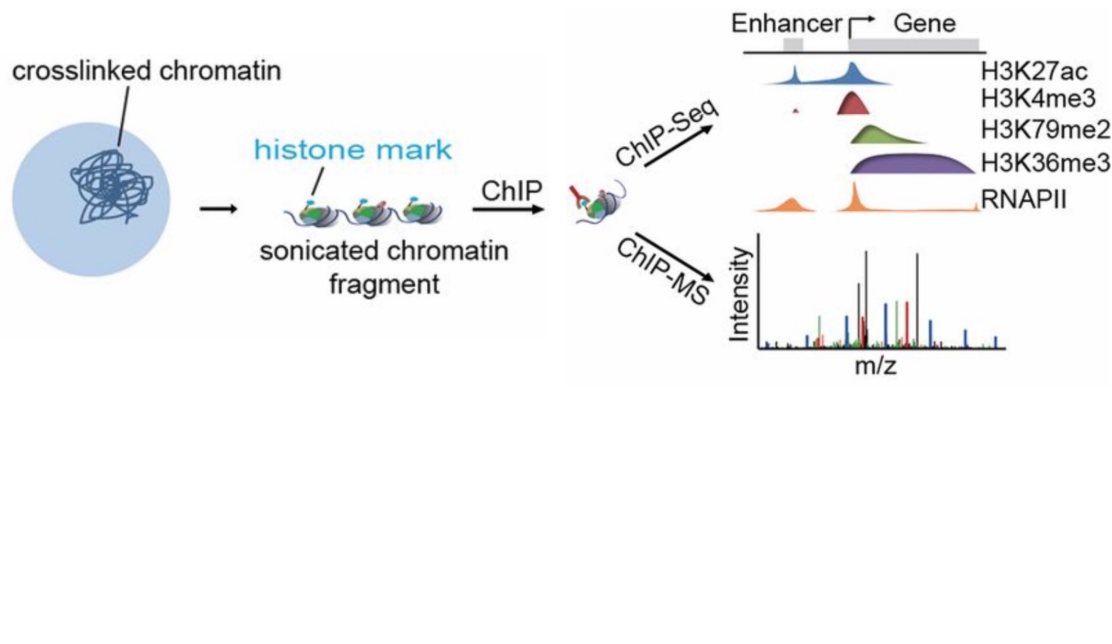


蛋白质质谱检测原理

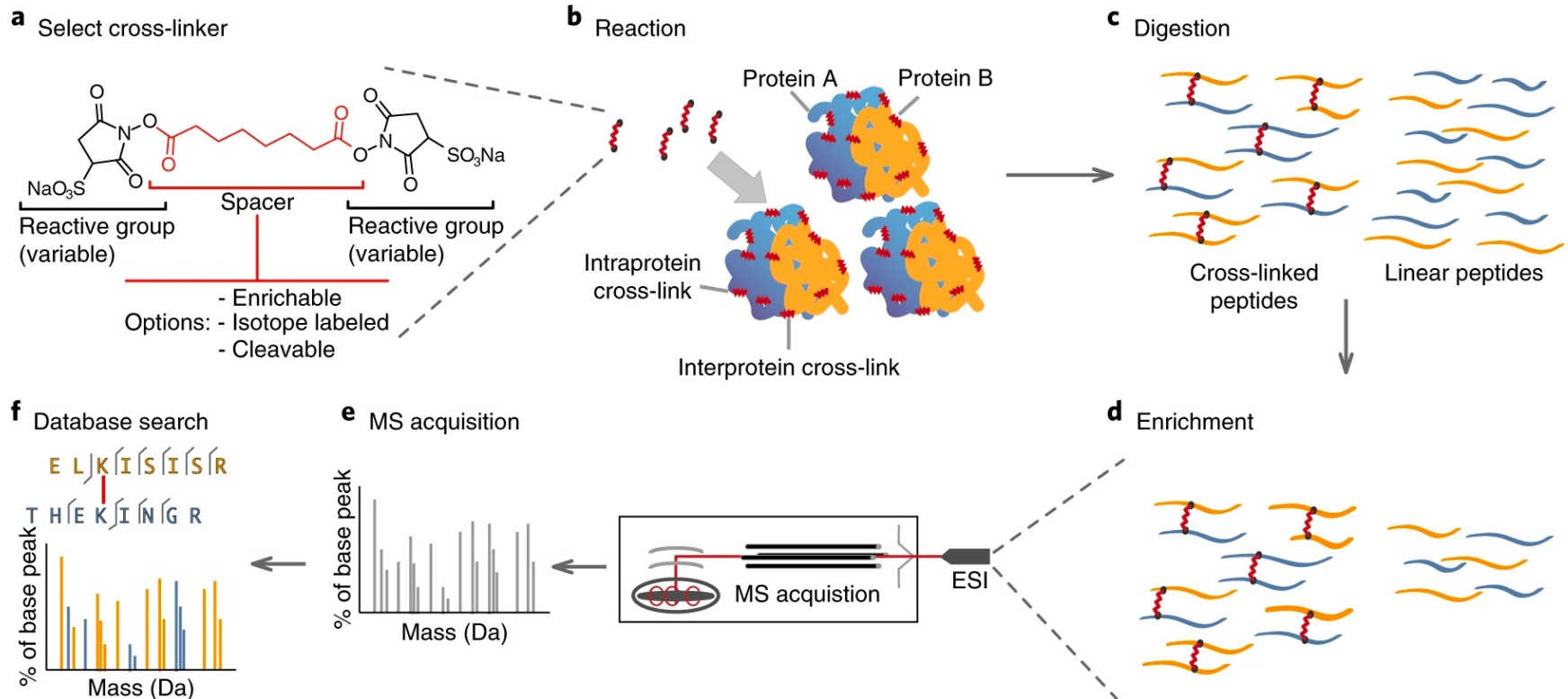


交联蛋白质组学

ChIP-MS



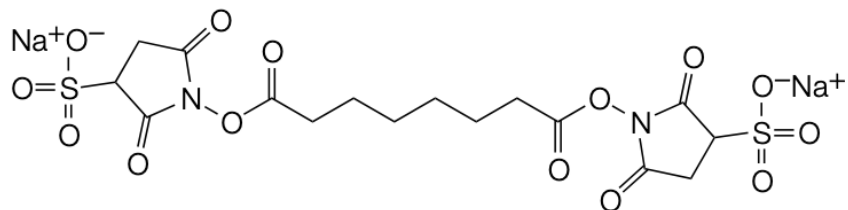
化学交联的蛋白质组分析技术



化学交联的蛋白质组分析技术

反应类别	靶点(靶标功能基团)	化学反应基团
氨基反应	-NH ₂	NHS酯 亚氨酸酯 五氟苯酯 羟甲基磷
羧基-氨基反应	-COOH	碳二亚胺(例如EDC)
巯基反应	-SH	马来酰亚胺 卤代乙酰基(溴-、氯-或碘-) 吡啶二巯基 硫代磺酸盐 乙烯基砷
醛基反应 (例如, 氧化糖、羰基)	-CHO	酰肼 烷氧基胺 NHS酯
光反应 (即无选择性、随机插入)	随机	双吡丙啶 芳香叠氮
羟基(非水性)-反应	-OH	异氰酸盐
叠氮-反应	-N ₃	炔烃 磷

化学交联剂的选择

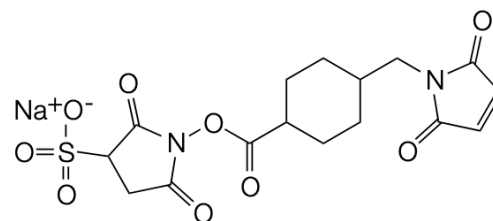


BS³

双(磺基琥珀酰亚氨基)辛二酸酯

MW 572.43

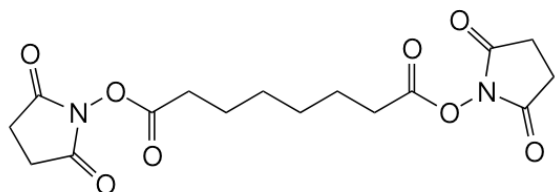
间隔臂 11.4 Å



Sulfo-SMCC, 磺基琥珀酰亚氨基4-(N-马来酰亚氨基甲基)环己烷-1-羧酸酯

MW 436.37

间隔臂 8.3 Å

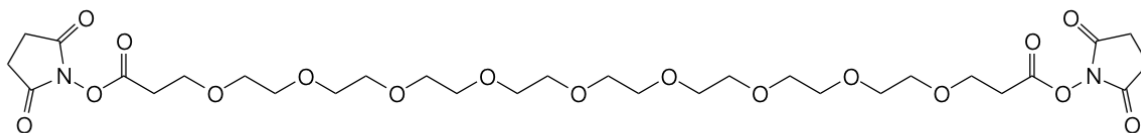


DSS

二琥珀酰亚氨基辛二酸酯

MW 368.34

间隔臂 11.4 Å



BS (PEG) 9

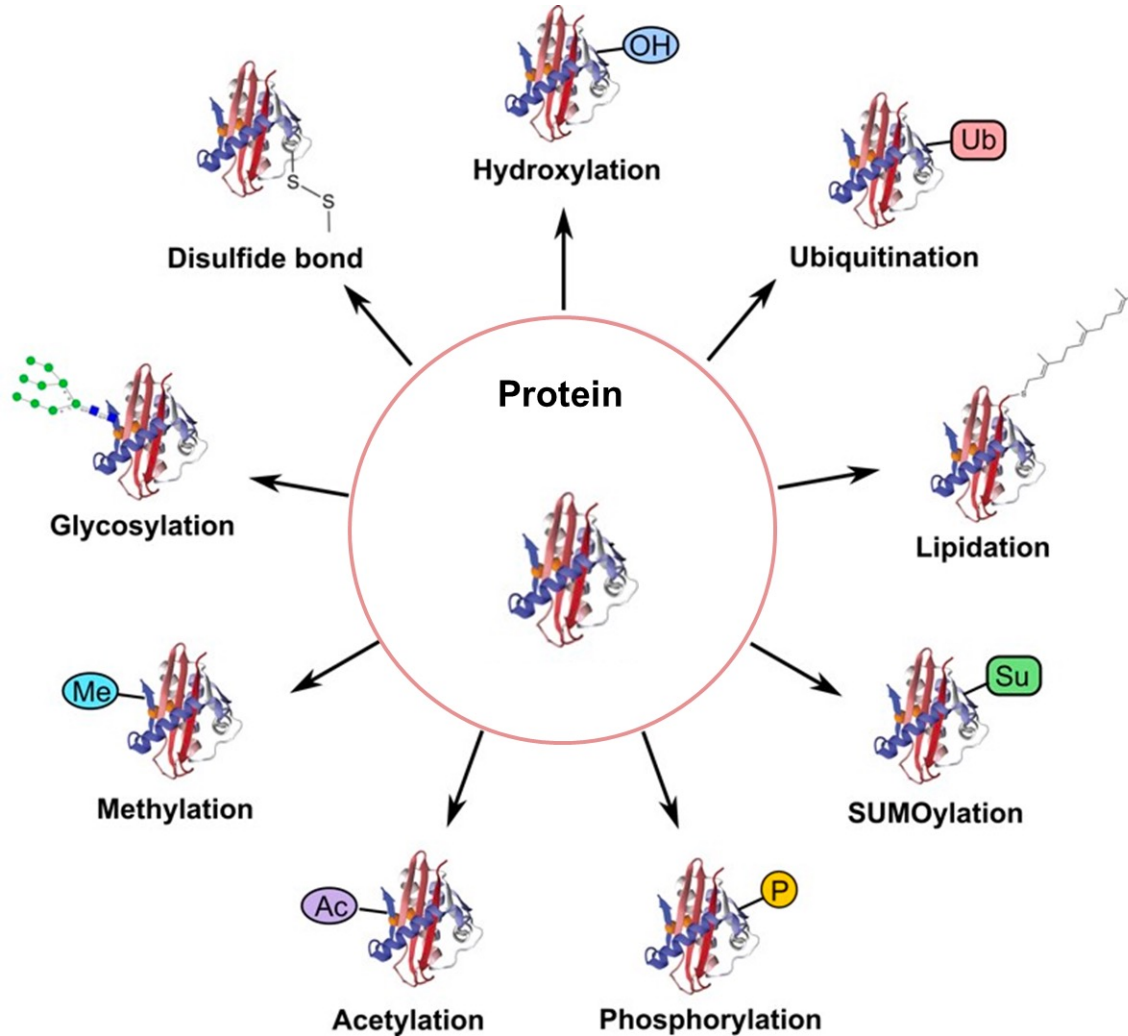
双(琥珀酰亚氨基)壬(乙二醇)

MW 708.71

间隔臂 35.8 Å

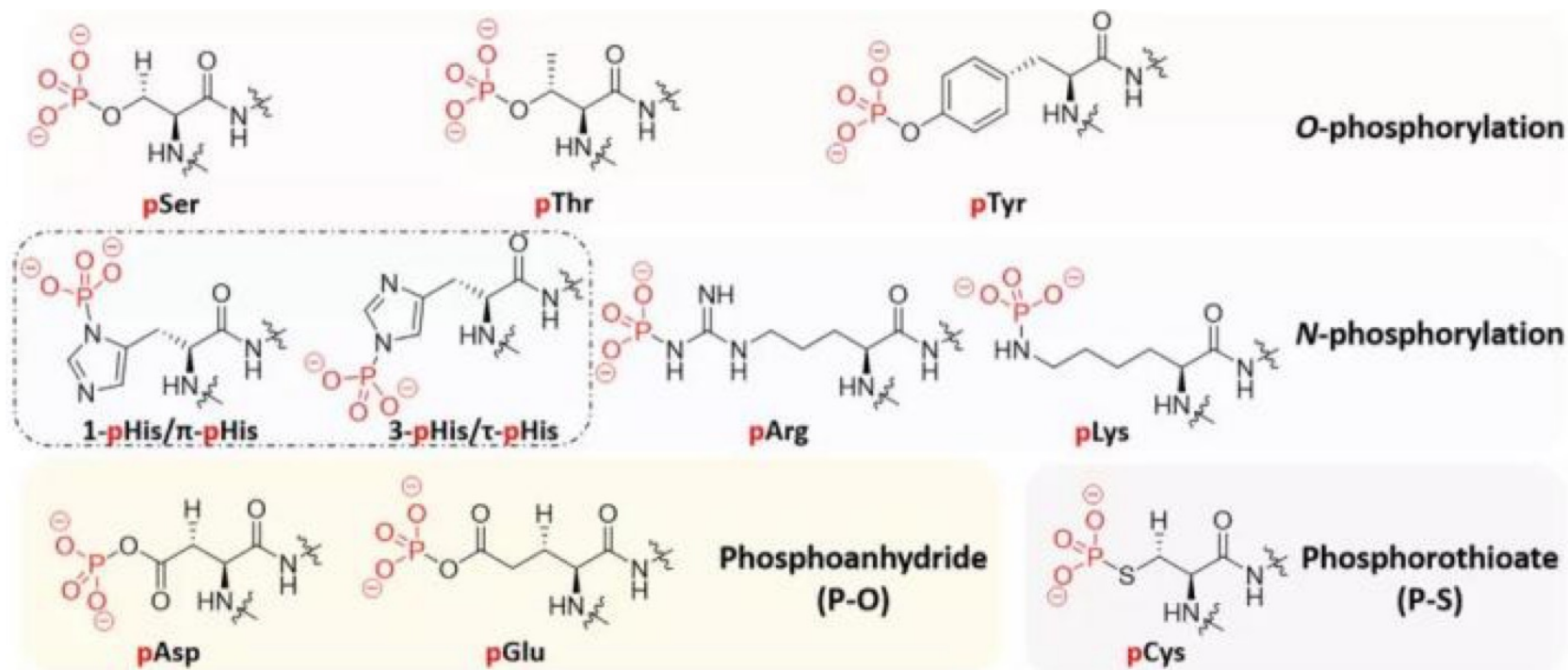
■ 课间休息

蛋白质的翻译后修饰



蛋白质翻译后修饰--磷酸化

■ 磷酸化



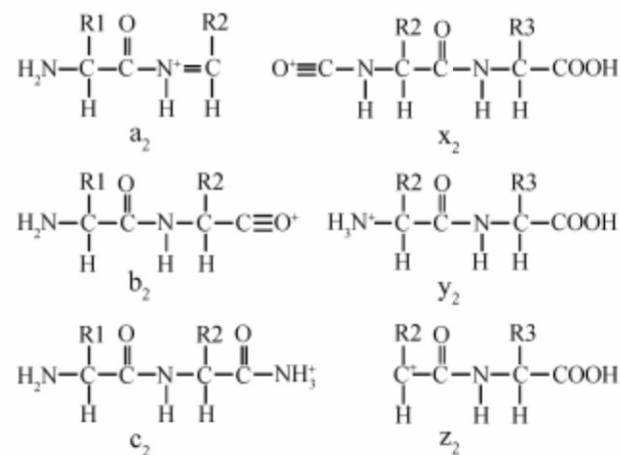
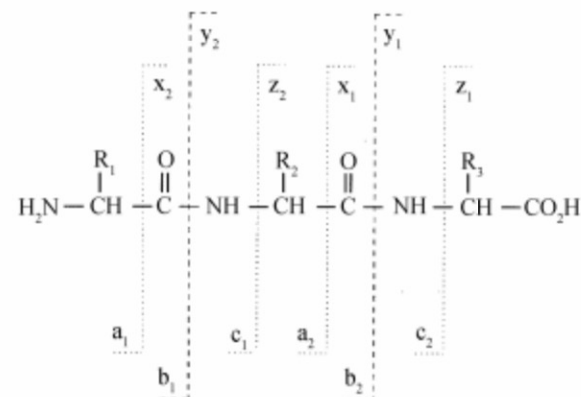
蛋白质翻译后修饰--磷酸化

蛋白质磷酸化类型	蛋白质磷酸化位点	氨基酸与磷酸基团间的 bond
O-磷酸化	丝氨酸 (Serine; Ser; S) ; 苏氨酸 (Threonine; Thr; T) ; 酪氨酸 (Tyrosine; Tyr; Y)	磷酸酯键 P-O
N-磷酸化	组氨酸 (Histidine; His; H) ; 精氨酸 (Arginine; Arg; R) ; 赖氨酸 (Lysine; Lys; K)	氨基磷酸酯键 P-N
S-磷酸化	天冬氨酸 (Aspartic acid; Asp; D); 谷氨酸 (Glutamic acid; Glu; E)	酸酐键 P-O
酰基磷酸化	半胱氨酸 (Cysteine; Cys; C)	硫代磷酸酯键 P-S

蛋白质翻译后修饰--磷酸化

磷酸化修饰的蛋白

- 改变所带的电荷
- 改变催化基团的性质改变蛋白质的构象
- 改变蛋白质的亚细胞分布
- 蛋白质功能变化的多样性

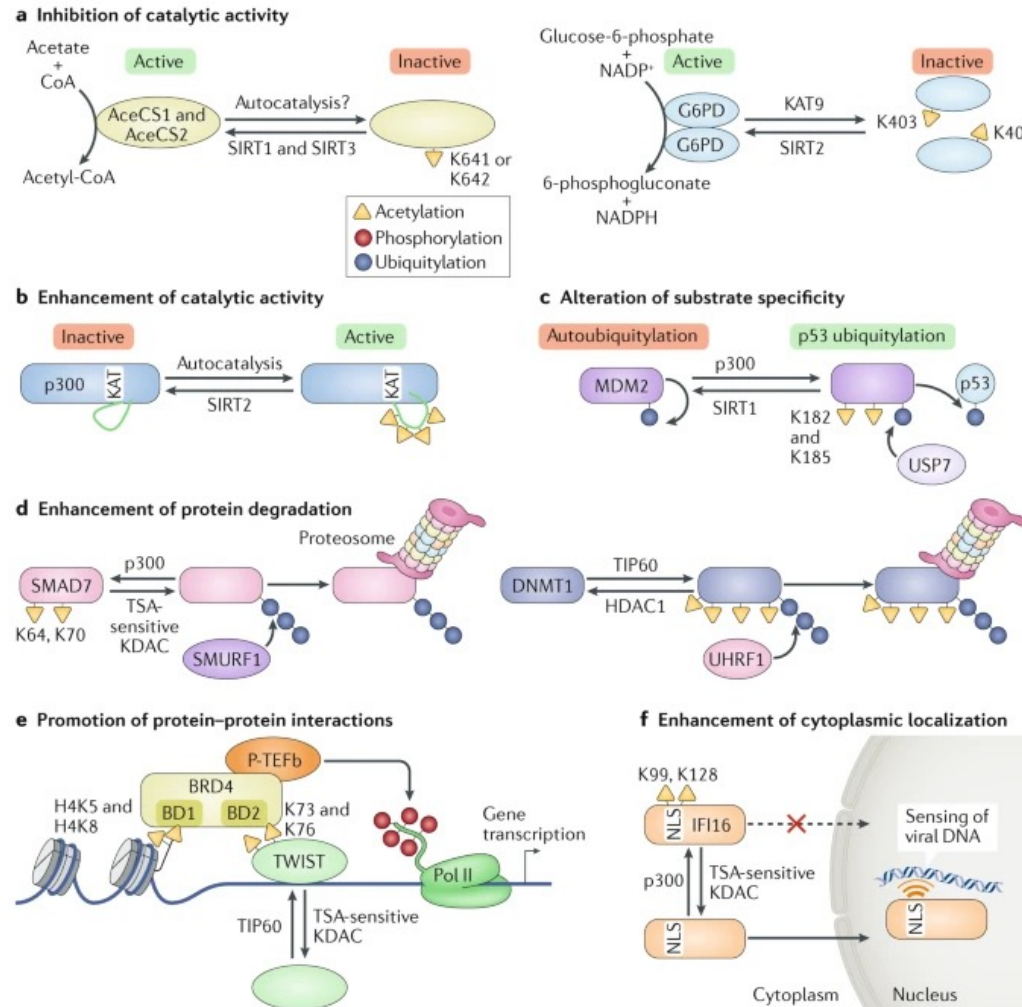


蛋白质翻译后修饰--磷酸化

磷酸肽的分离和富集方法

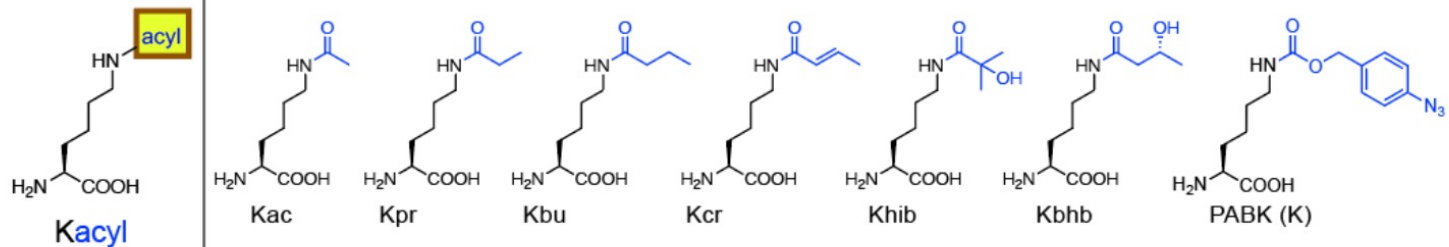
- 反相液相色谱(PR-HPLC)
- 免疫沉淀
- 固相金属亲和色谱
- 金属氧化物/氢氧化物亲和色谱
- 离子交换和等电聚焦

蛋白质翻译后修饰—乙酰化



Linking chromatin acylation mark-defined proteome and genome in living cells

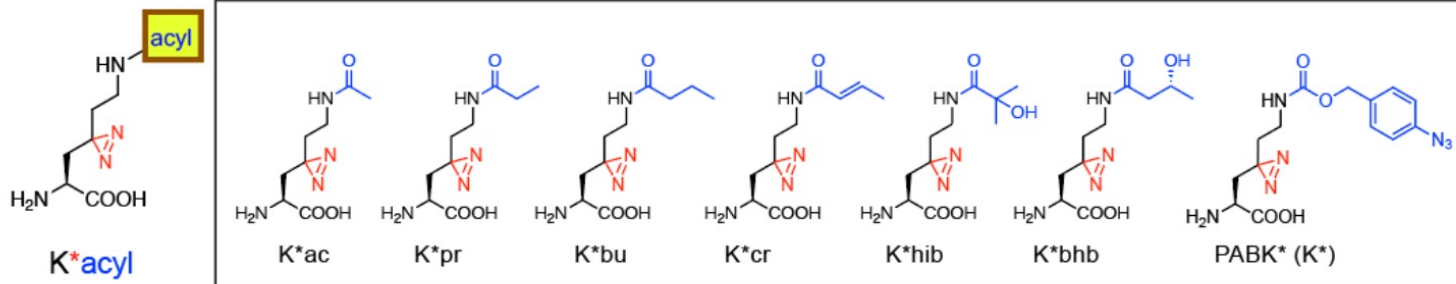
SiTomics



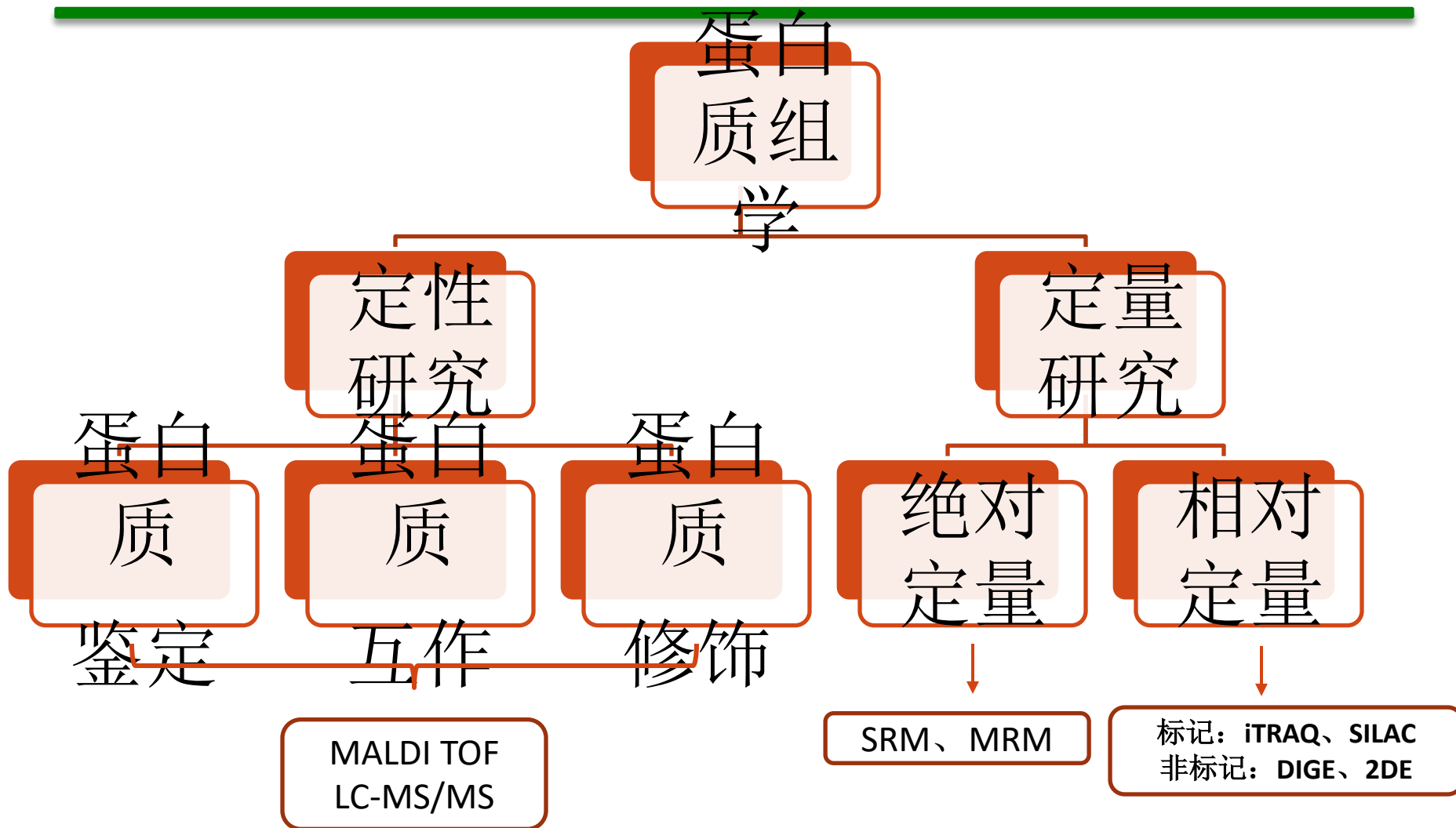
遗传密码子拓展策略



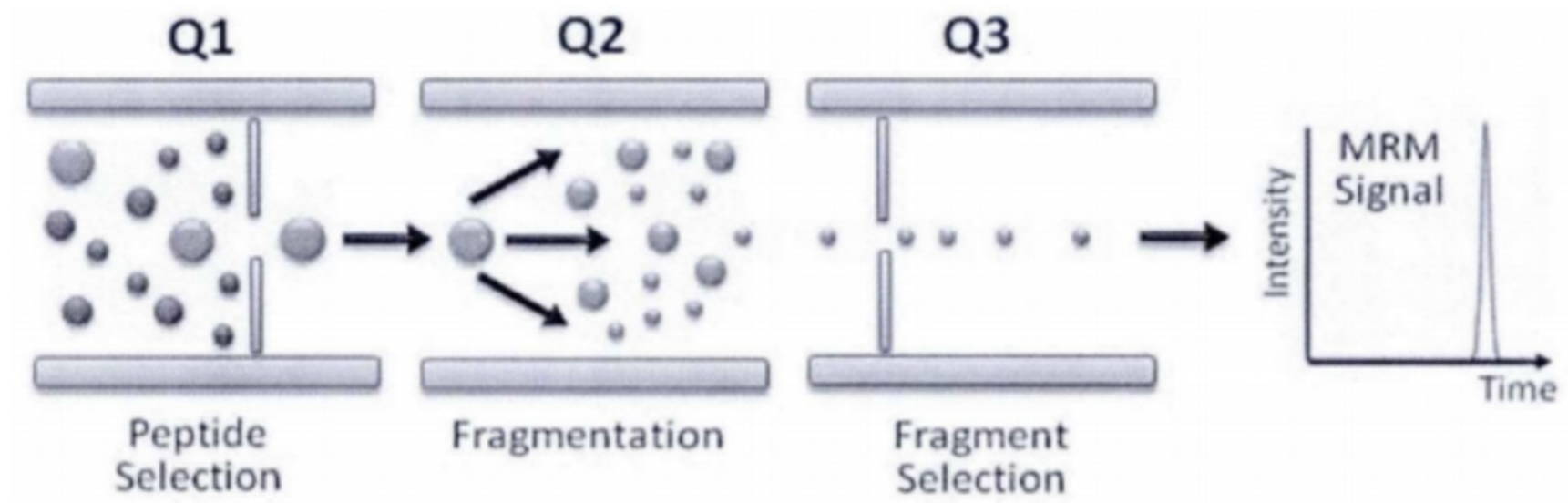
Photoaffinity



定量蛋白质组学研究技术



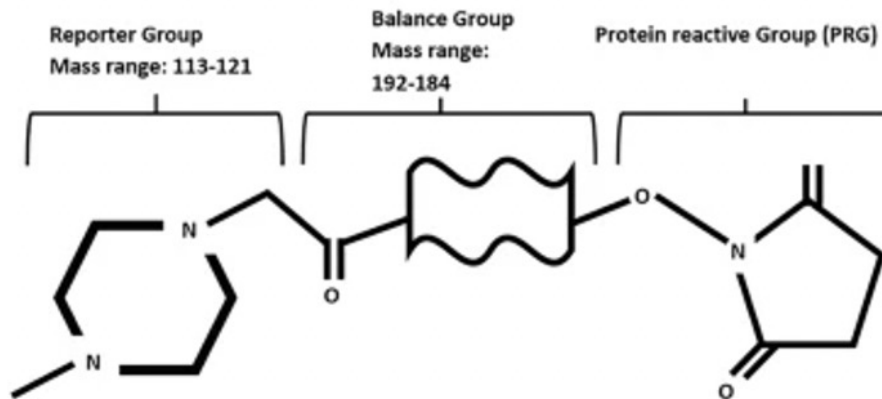
蛋白质组绝对定量技术（MRM/SRM, PRM）



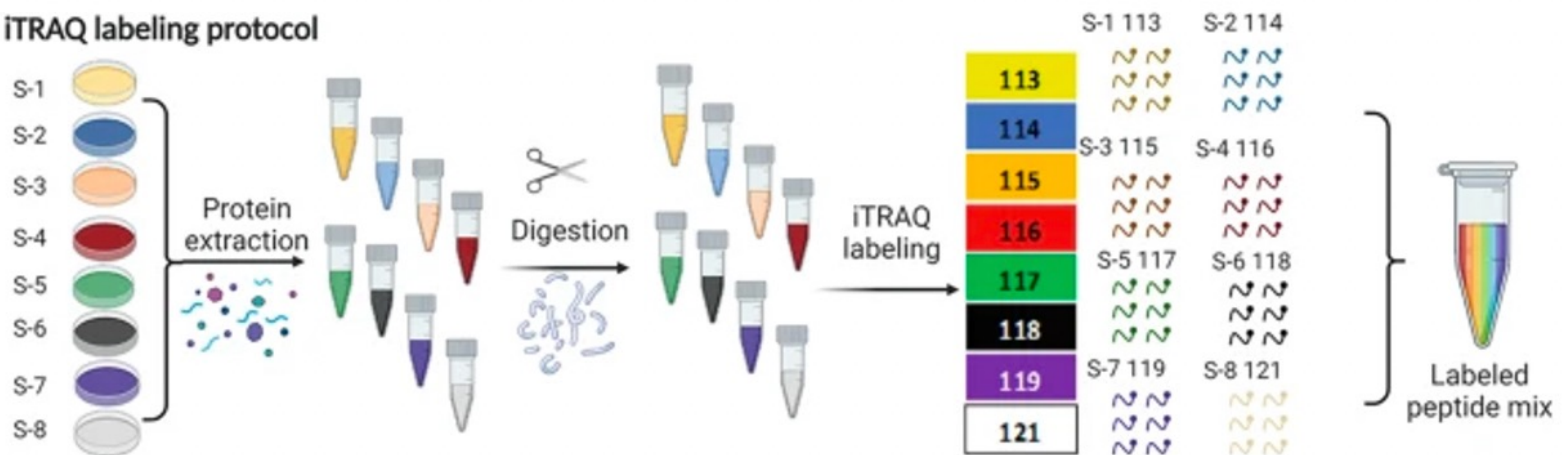
标准品

蛋白质组相对定量技术 (iTRAQ)

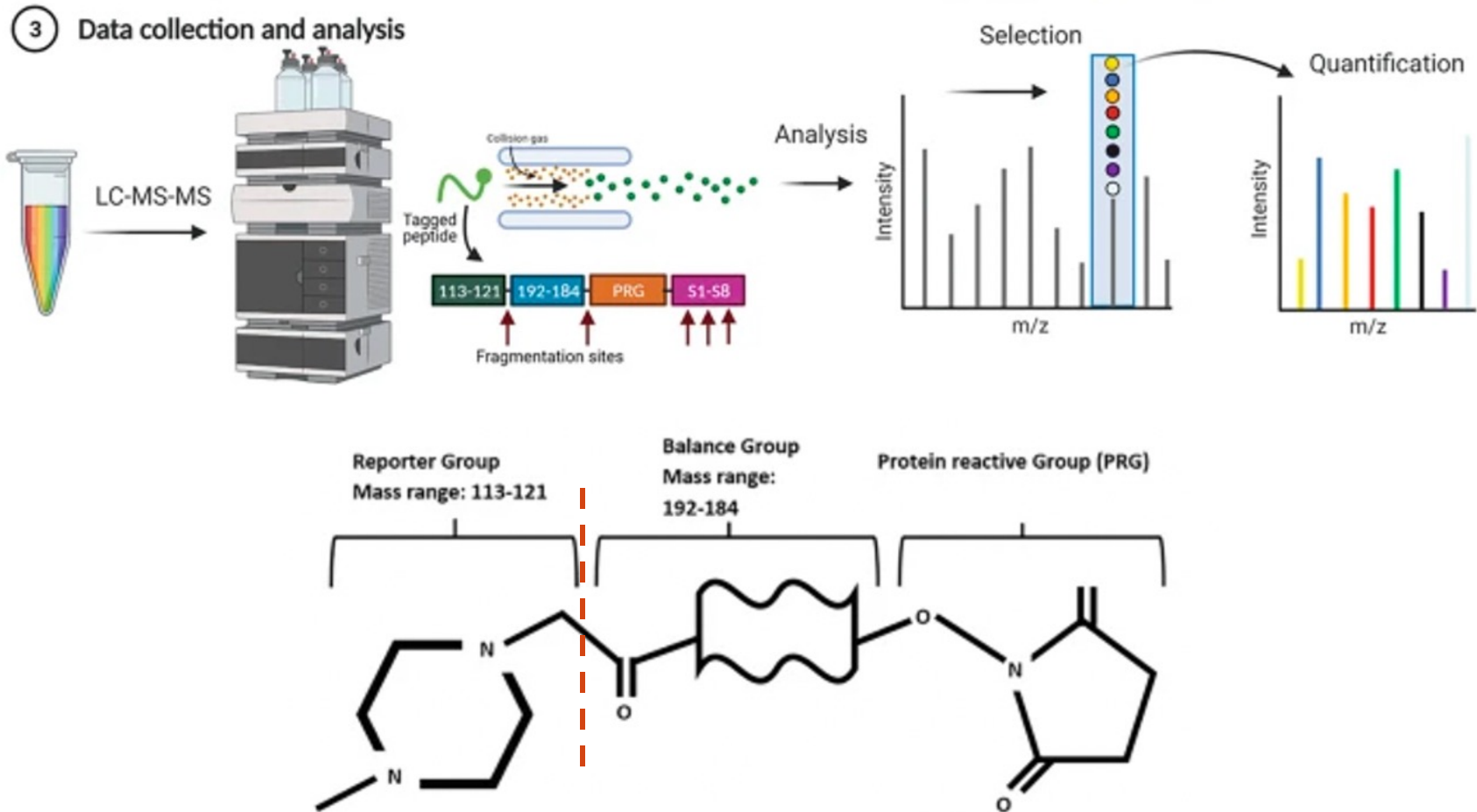
① iTRAQ chemical structure and design



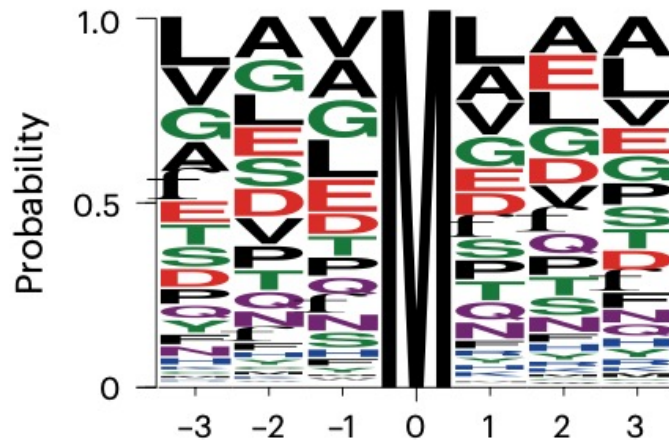
② iTRAQ labeling protocol



蛋白质组相对定量技术 (iTRAQ)



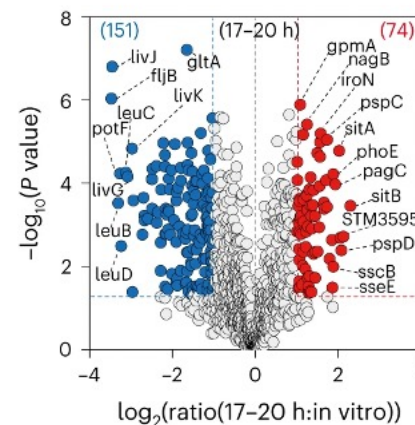
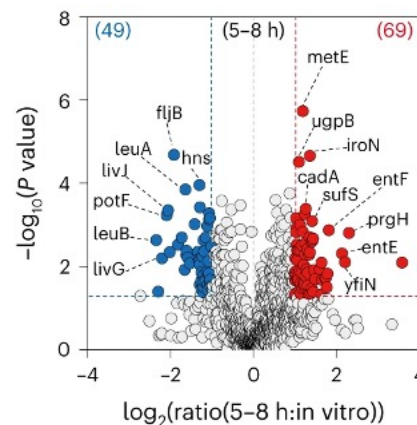
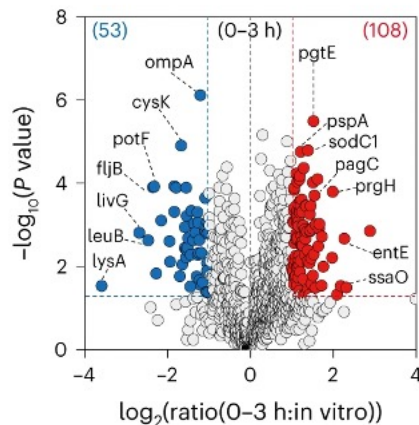
蛋白质组研究中的生物信息学



1) motif

<http://weblogo.threeplusone.com>

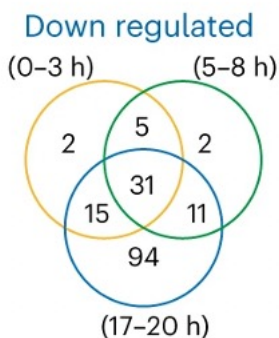
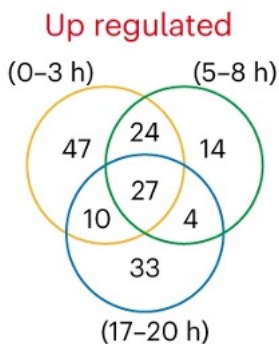
MEME



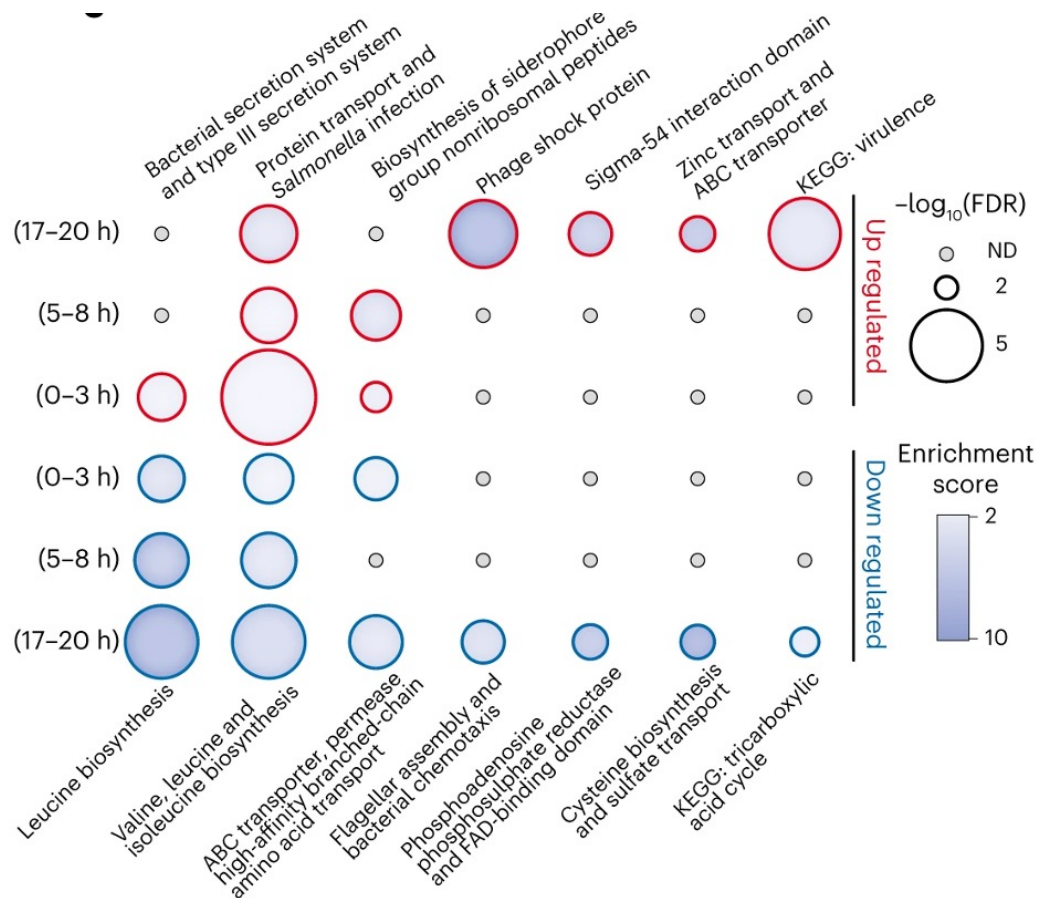
2) 差异表达分析

蛋白质组研究中的生物信息学

3) 韦恩图

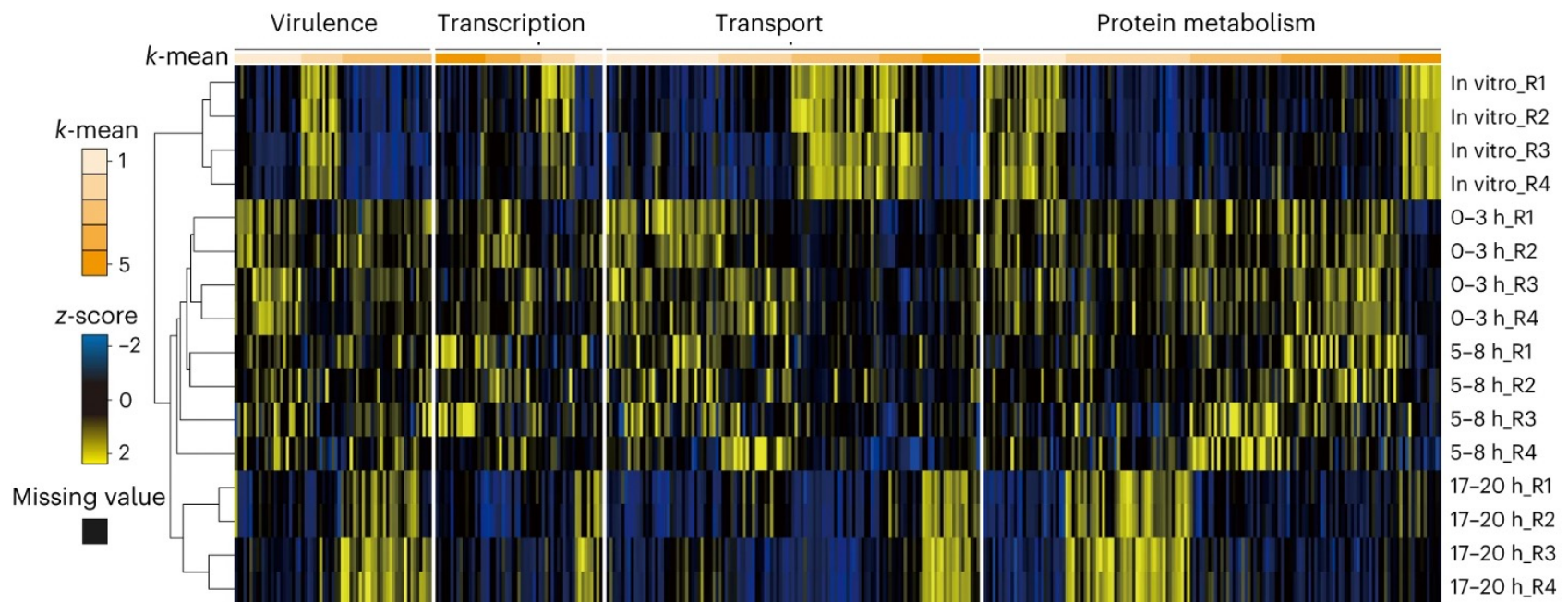


4) 气泡图 富集分析

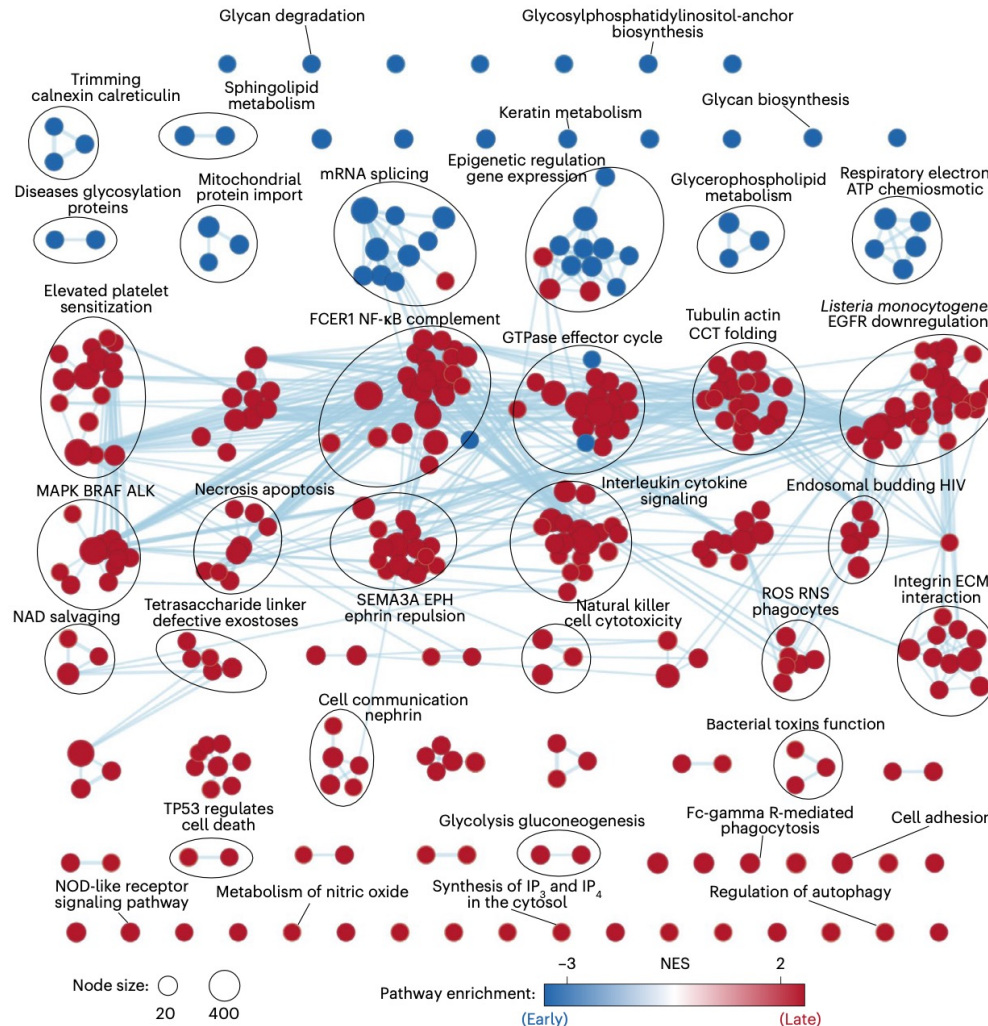


蛋白质组研究中的生物信息学

5) 热图 富集分析



蛋白质组研究中的生物信息学



6) PPI分析

STRING: functional protein association networks

 **STRING**

SearchDownloadHelpMy Data

Protein by name >
Multiple proteins >
Proteins by sequences >
Proteins with Values/Ranks >
Protein families ("COGs") >
Pathway / Process / Disease ^{New} >
Annotate your proteome ^{New} >
Organisms >
Examples >
Random entry >

SEARCH

Multiple Proteins by Names / Identifiers

List Of Names: (one-per-line or CSV; examples: #1 #2 #3)

ACBD3
ACTA1
ACTA2
ACTB
ACTC1
AIP

... or, upload a file:

Browse ...

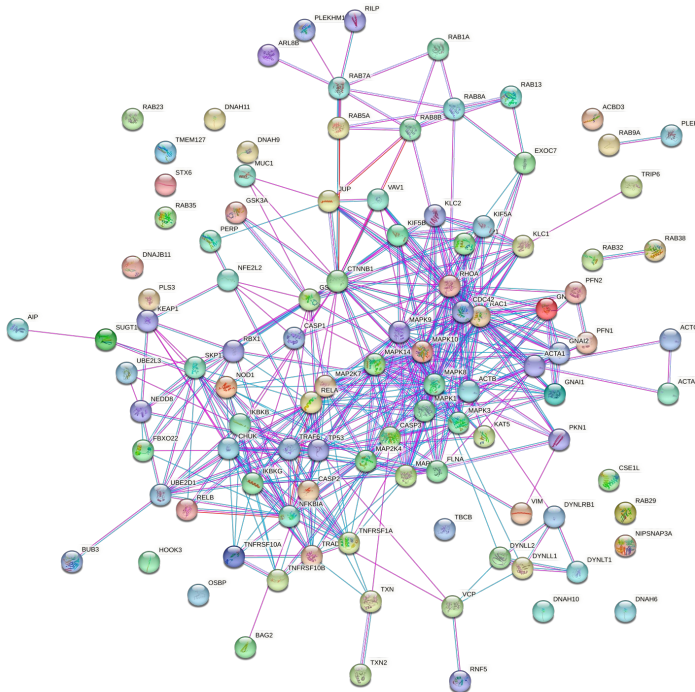
Organisms:

auto-detect  

[Advanced Settings](#)

SEARCH

STRING: functional protein association networks



Viewers >

Legend v

Settings >

Analysis >

Exports >

Clusters >

+ More

- Less

Nodes:

Network nodes represent proteins
splice isoforms or post-translational modifications are collapsed, i.e. each node represents all the proteins produced by a single, protein-coding gene locus.

Node Color

colored nodes:
query proteins and first shell of interactors

white nodes:
second shell of interactors

Node Content

empty nodes:
proteins of unknown 3D structure

filled nodes:
some 3D structure is known or predicted

Edges:

Edges represent protein-protein associations
associations are meant to be specific and meaningful, i.e. proteins jointly contribute to a shared function; this does not necessarily mean they are physically binding to each other.

Known Interactions

from curated databases

experimentally determined

Predicted Interactions

gene neighborhood

gene fusions

gene co-occurrence

Others

textmining

co-expression

protein homology

Your Input:

GNAI3

Guanine nucleotide-binding protein G(k) subunit alpha; Heterotrimeric guanine nucleotide-binding proteins (G proteins) function as transducers downstream of G protein-coupled receptors (GPCRs) in numerous signaling cascades. The alpha chain contains the guanine nucleotide binding site and alternates between an active, GTP-bound state and an inactive, GDP-bound state. Signaling by an activated GPCR promotes GDP release and GTP binding. The alpha subunit has a low GTPase activity that converts bound GTP to GDP, thereby terminating the signal. Both GDP release and GTP hydrolysis are modul [...] (354 aa)

NOD1

Nucleotide-binding oligomerization domain-containing protein 1; Enhances caspase-9-mediated apoptosis. Induces NF-kappa-B activity via RIPK2 and IKK-gamma. Confers responsiveness to intracellular bacterial lipopolysaccharides (LPS). Forms an intracellular sensing system along with ARHGAP2 for the detection of microbial effectors during cell invasion by pathogens. Required for BLAA and BIK3 dependent NF-kappa-B

42

STRING: functional protein association networks

Viewers > Legend > Settings > Analysis > Exports > Clusters > More > Less

Network Stats

number of nodes: 105
number of edges: 410
average node degree: 7.81
avg. local clustering coefficient: 0.501

expected number of edges: 141
PPI enrichment p-value: < 1.0e-16
your network has significantly more interactions
than expected (*what does that mean?*)

Functional enrichments in your network

Note: some enrichments may be expected here (*why?*)

[explain columns](#)

Biological Process (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:2000077	Negative regulation of type B pancreatic cell development	2 of 2	2.27	0.0040
GO:2000466	Negative regulation of glycogen (starch) synthase activity	2 of 3	2.09	0.0062
GO:0090131	Mesenchyme migration	3 of 5	2.05	0.00030
GO:0048210	Golgi vesicle fusion to target membrane	3 of 5	2.05	0.00030
GO:0035617	Stress granule disassembly	3 of 5	2.05	0.00030
(more ...)				

Molecular Function (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0004705	JUN kinase activity	3 of 3	2.27	0.00017
GO:0036461	BLOC-2 complex binding	2 of 2	2.27	0.0073
GO:0008545	JUN kinase kinase activity	2 of 2	2.27	0.0073
GO:0051022	Rho GDP-dissociation inhibitor binding	2 of 3	2.09	0.0113
GO:0035651	AP-3 adaptor complex binding	2 of 3	2.09	0.0113
(more ...)				

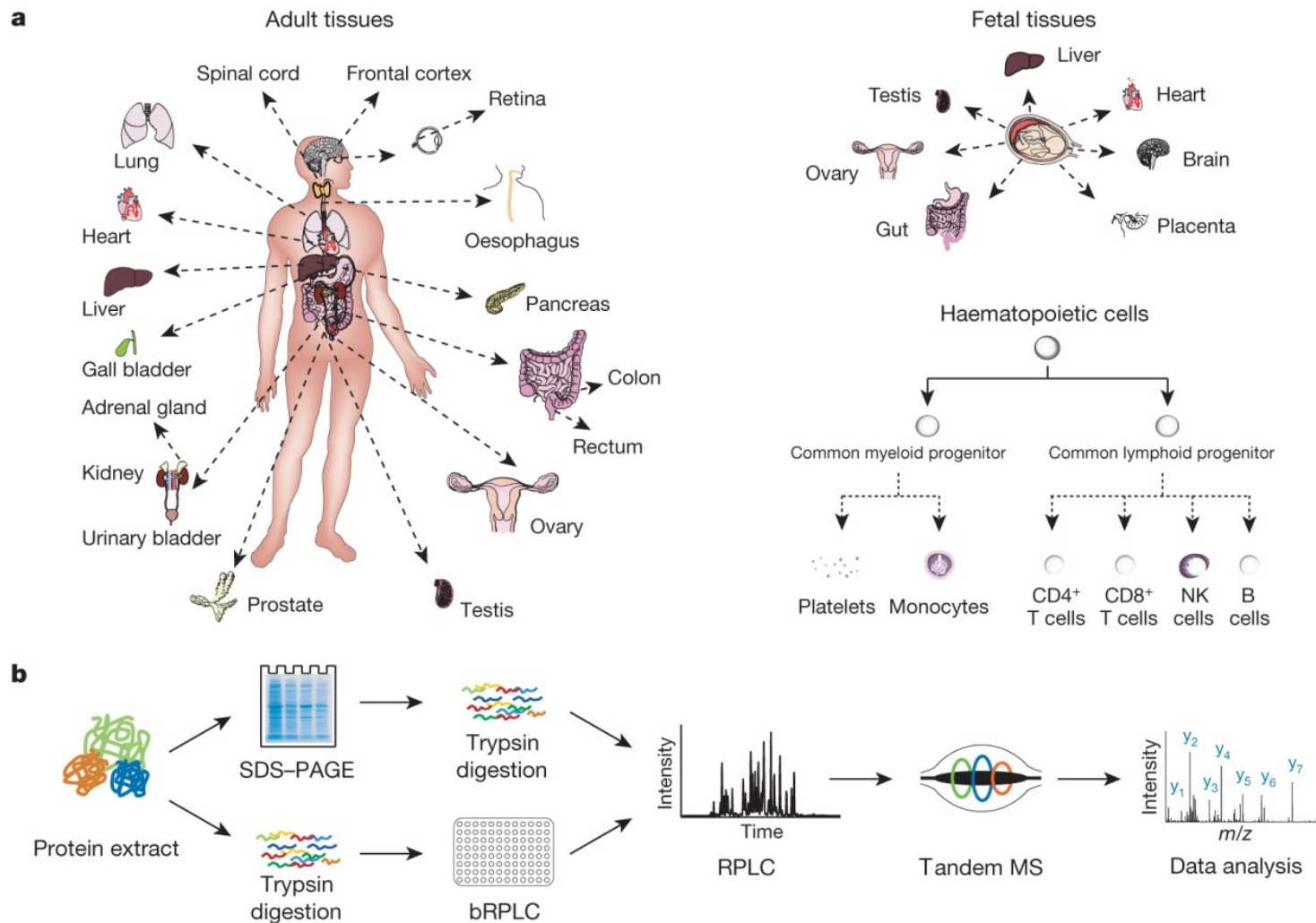
Cellular Component (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0071664	catenin-TCF7L2 complex	2 of 4	1.97	0.0068
GO:0031467	Cul7-RING ubiquitin ligase complex	2 of 4	1.97	0.0068
GO:0002947	Tumor necrosis factor receptor superfamily complex	2 of 4	1.97	0.0068
GO:0033256	I-kappaB/NF-kappaB complex	3 of 7	1.9	0.00039
GO:0008385	IkappaB kinase complex	3 of 7	1.9	0.00039
(more ...)				

Reference publications (PubMed)				
publication	(year) title	count in network	strength	false discovery rate
PMID:9054391	(1997) Selective activation of caspases during apoptotic in...	6 of 6	2.27	8.47e-08
PMID:22740410	(2012) HTLV-1 tax-induced rapid senescence is driven by th...	6 of 6	2.27	8.47e-08
PMID:30506715	(2019) Shared KEGG pathways of icariin-targeted genes and...	5 of 5	2.27	1.86e-06
PMID:28533220	(2017) Triptolide-Assisted Phosphorylation of p53 Suppress...	5 of 5	2.27	1.86e-06
PMID:21816815	(2011) Disruption of IkappaB kinase (IKK)-mediated RelA se...	5 of 5	2.27	1.86e-06
(more ...)				

Save / Export

Biological Process (Gene Ontology)	download	705 GO-terms sign
Molecular Function (Gene Ontology)	download	74 GO-terms sign
Cellular Component (Gene Ontology)	download	124 GO-terms sign
Reference publications (PubMed)	download	10000 publication
Local network cluster (STRING)	download	29 clusters signi
KEGG Pathways	download	153 pathways sign
Reactome Pathways	download	233 pathways sign
WikiPathways	download	216 pathways sign
Tissue expression (TISSUES)	download	71 tissues signifi
Subcellular localization (COMPARTMENTS)	download	95 compartment
Human Phenotype (Monarch)	download	36 phenotypes si
Annotated Keywords (UniProt)	download	33 keywords sign
Protein Domains (Pfam)	download	25 domains signi
Protein Domains and Features (InterPro)	download	62 domains signi
Protein Domains (SMART)	download	one single doman
All enriched terms (without PubMed)	download	1857 enriched te
Selected terms only	download	no enriched term

人类蛋白质组草图



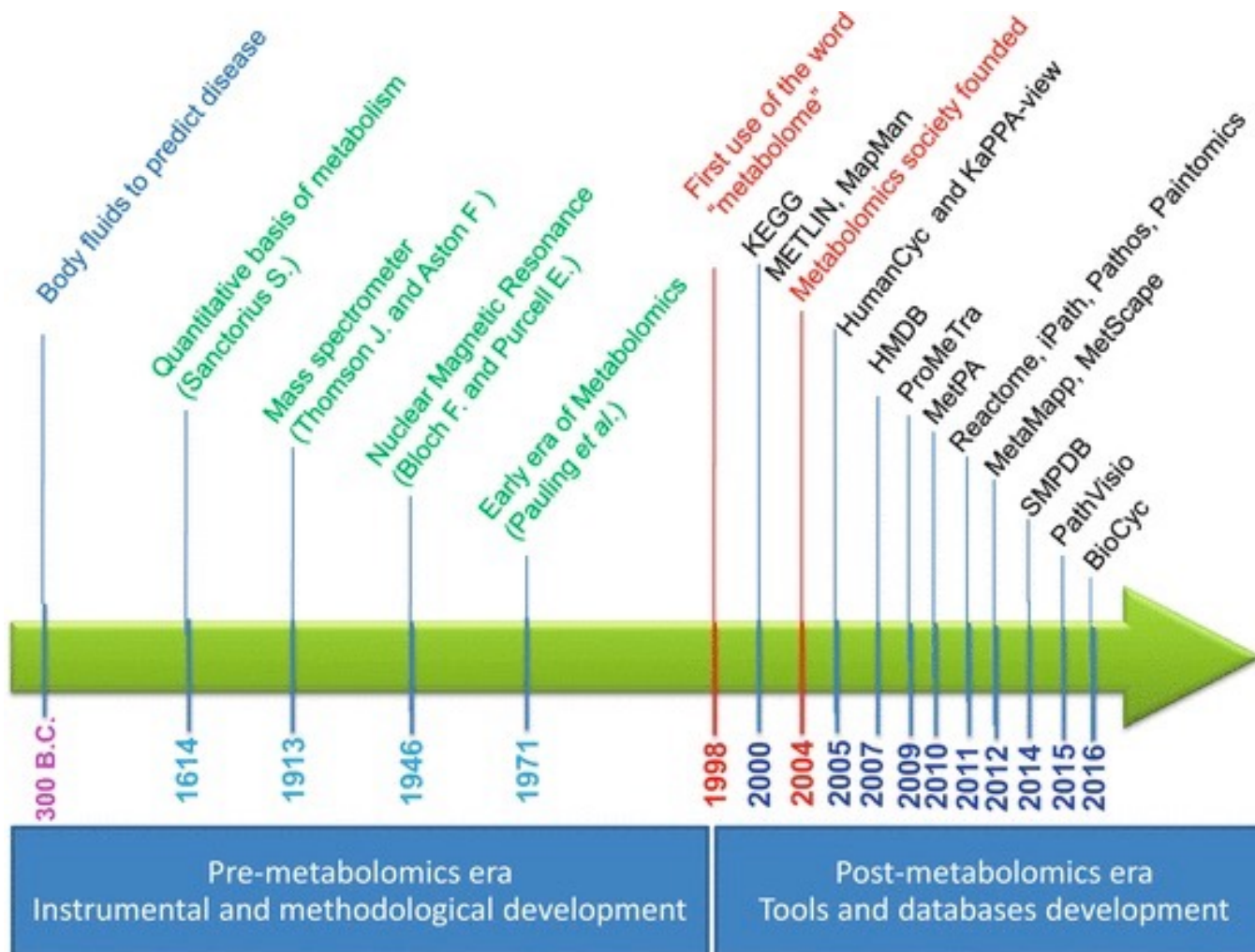
代谢组

Metabonomics of yore

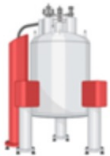


This urine wheel was published in 1506 by Ullrich Pinder, in his book *Epiphania Medicorum*. It describes the possible colours, smells and tastes of urine, and uses them to diagnose disease.

代谢组学发展历史

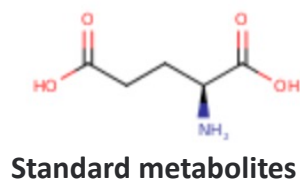


代谢组学研究技术

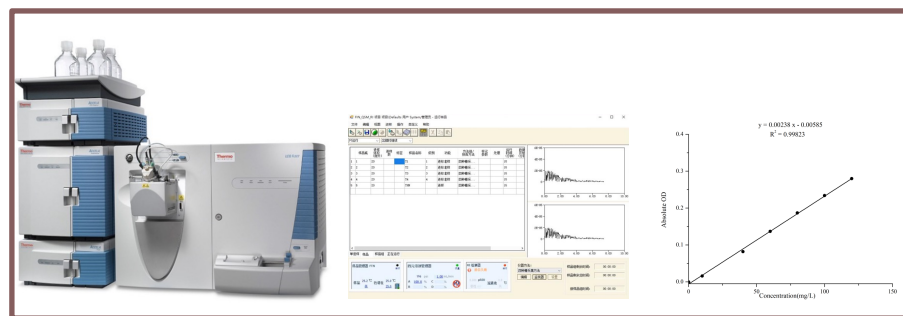
Analytical Devices	Scientific Instruments	Key Functions	Metabolic Applications	Ref
MS		High throughput, high sensitivity/resolution, capable of quickly examining, reduces complexity, improves resolution, specificity, and quantification, allows for isotopic labeling, offers structural information, performed under ambient environmental conditions with the preservation of tissue morphology (DESI-MS).	GC-MS: SCFAs and ketones, carbohydrate metabolites, amino acids. LC-MS: amino acids and their byproducts, bile acids, lipids and fatty acids, sugar metabolites, vitamins, and related compounds. Imaging MS: MALDI/DESI-IMS and nano SIMS.	[31,32,33,34,36,37,38,43,47].
NMR		Affords structural evidence, low-sensitivity compared to MS, high-throughput, permits the quantification of isotopic labeling, delivers spatial data (NMR imaging or MRI).	Sugar metabolites, amino acids and amino acid byproducts SCFAs, vitamins, untargeted analysis, and metabolome finger printing.	[22,23,44,45,48]
Raman MS		3D info, high-throughput, structural data, non-destructive methods, lower sensitivity versus MS and NMR.	This can be united with fluorescent probes and isotopic labeling for the single-cell-resolved assessment of nutrient assimilation.	[43]
UHPLC		High-sensitivity detection	Detection and identification of a broad range of metabolites	[49,50]
Immunochemistry and enzymatic assays	--	Low-throughput, high specificity, may provide spatial information (immunohistochemistry or immunofluorescence).	Eicosanoids, uric acid, serotonin, neurotransmitters, lipopolysaccharide, some vitamins, sugar metabolites.	[51,52]

代谢组学分析策略

■ 代谢物靶标分析 (Metabolite Target Analysis) :对个别特定组分分析



Methods



Extract

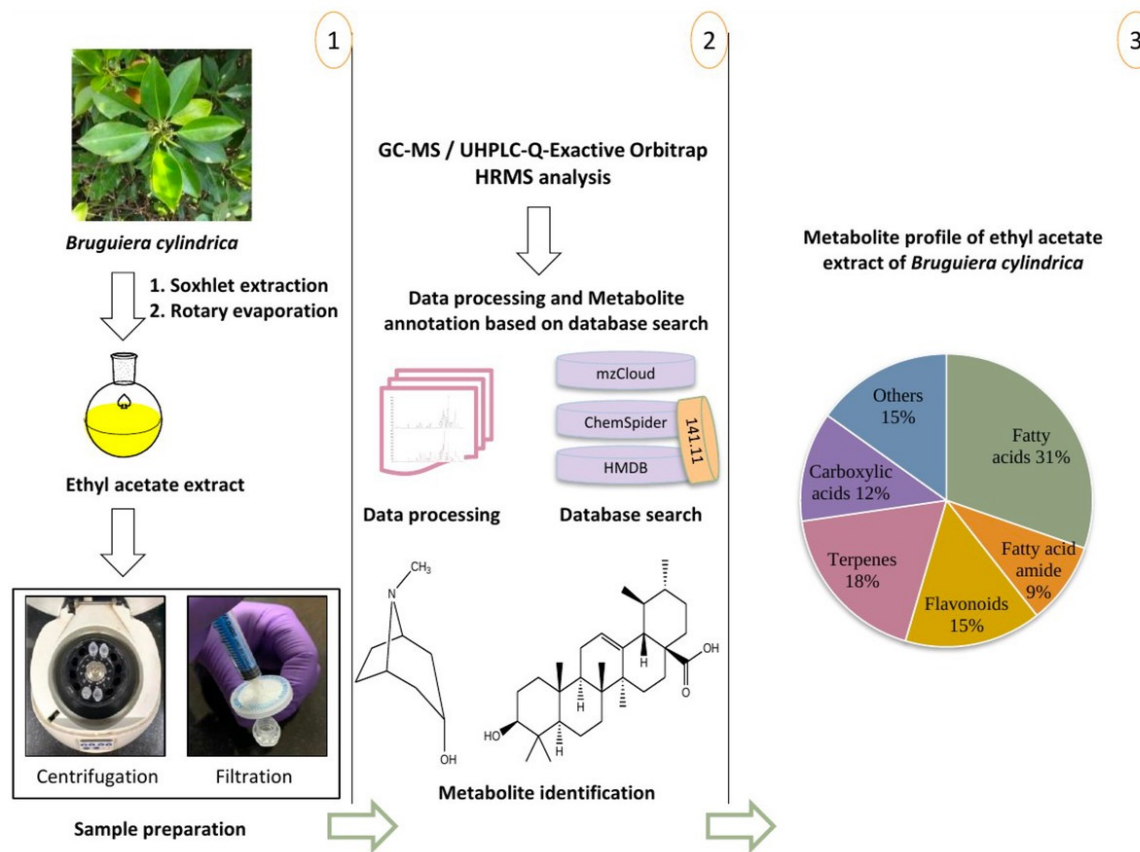


定量分析

Chromatography-MS/MS

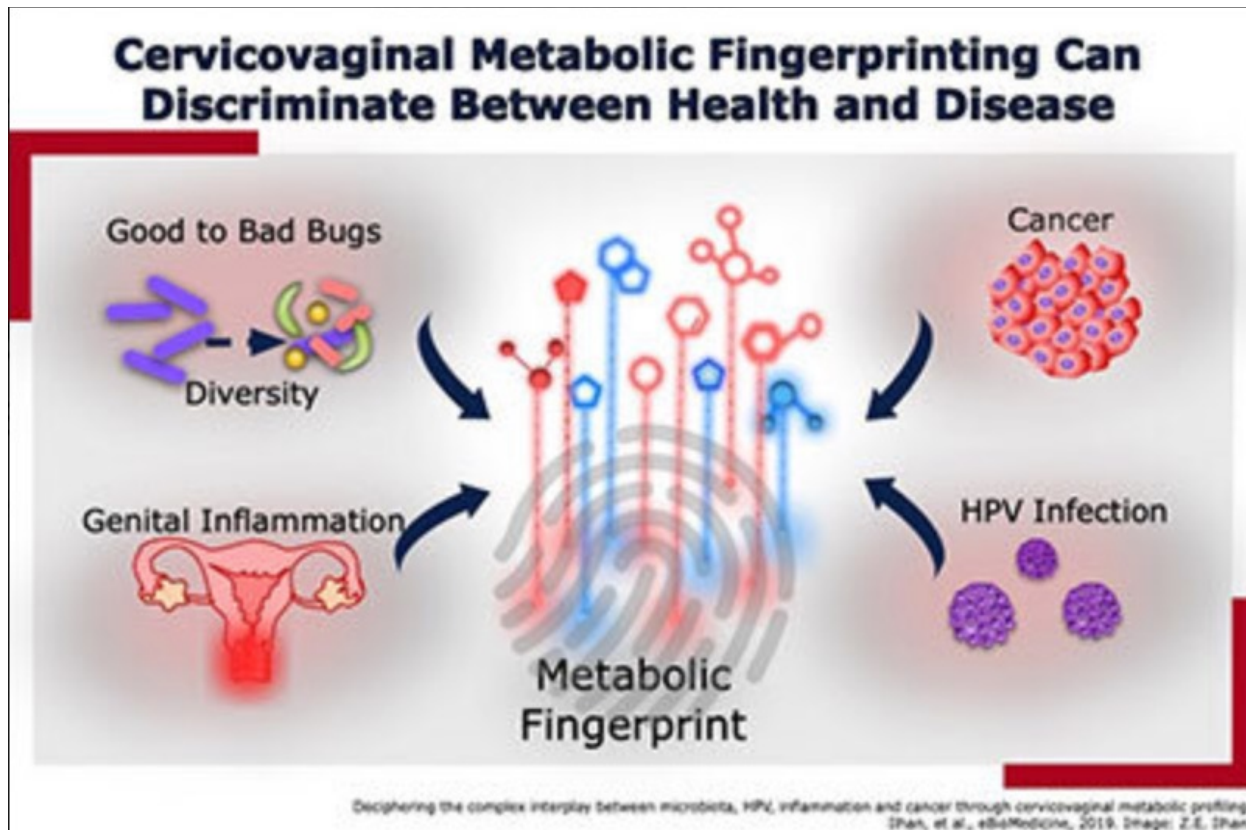
代谢组学分析策略

■ 代谢组学:特定样品中所有代谢物分析



代谢组学分析策略

■ 代谢指纹分析: 比较代谢物指纹图谱



代谢组学数据分析

■ 数据质控和标准化处理

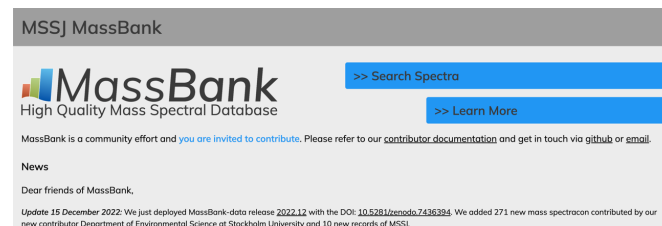
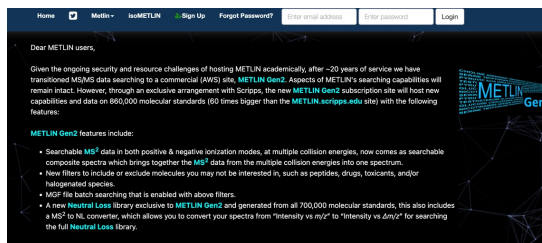
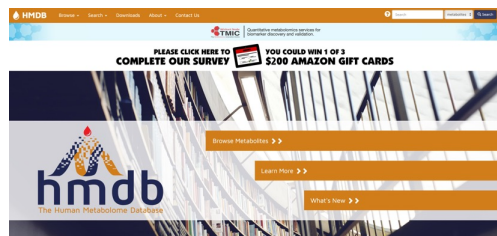
■ 原始数据的处理

降噪、谱图解卷积、峰检测和积分、色谱图对齐、化合物识别和定量。

■ 数据分析

PCA 分析、热图、火山图、通路分析

■ 常用数据库网站



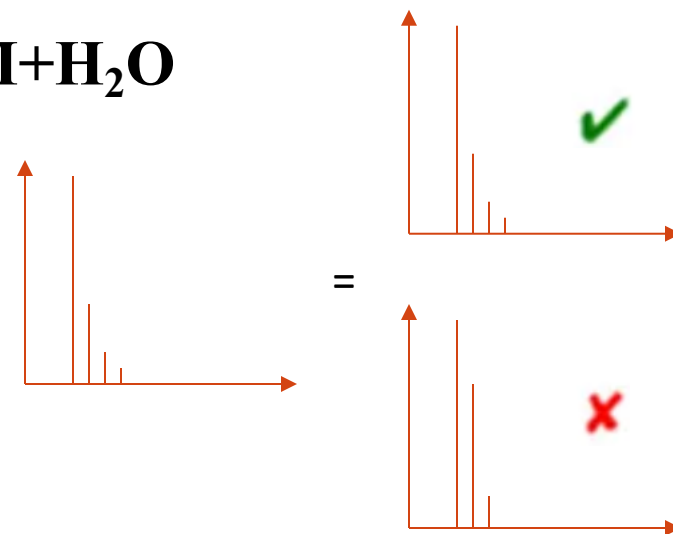
代谢产物的鉴定

■ 一级鉴定：解决质量数问题

1) $M+H$ 、 $M+Na$ 、 $M+K$ 、 $M+NH_4^+$ 、 $M+H_2O$

2)

$^{12}C : ^{13}C$ 100 : 1.08	$^{16}O : ^{17}O : ^{18}O$ 100 : 0.04 : 0.20	$^{35}Cl : ^{37}Cl$ 100 : 32.5
$^1H : ^2H$ 100 : 0.016	$^{28}Si : ^{29}Si : ^{30}Si$ 100 : 5.1 : 3.35	$^{79}Br : ^{81}Br$ 100 : 98.0
$^{14}N : ^{15}N$ 100 : 0.38	$^{32}S : ^{33}S : ^{34}S$ 100 : 0.78 : 4.40	



■ 二级鉴定：解决同分异构体问题

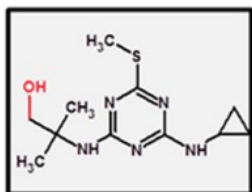
多级质谱+其他谱图

代谢产物的鉴定

Example

Identification confidence

Minimum data requirements

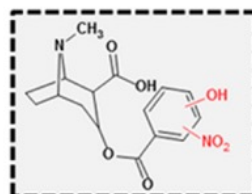


Level 1: Confirmed structure
by reference standard

MS, MS^2 , RT, Reference Std.

Level 2: Probable structure
a) by library spectrum match
b) by diagnostic evidence

MS, MS^2 , Library MS^2
MS, MS^2 , Exp. data



Level 3: Tentative candidate(s)
structure, substituent, class

MS, MS^2 , Exp. data

$\text{C}_6\text{H}_5\text{N}_3\text{O}_4$

Level 4: Unequivocal molecular formula

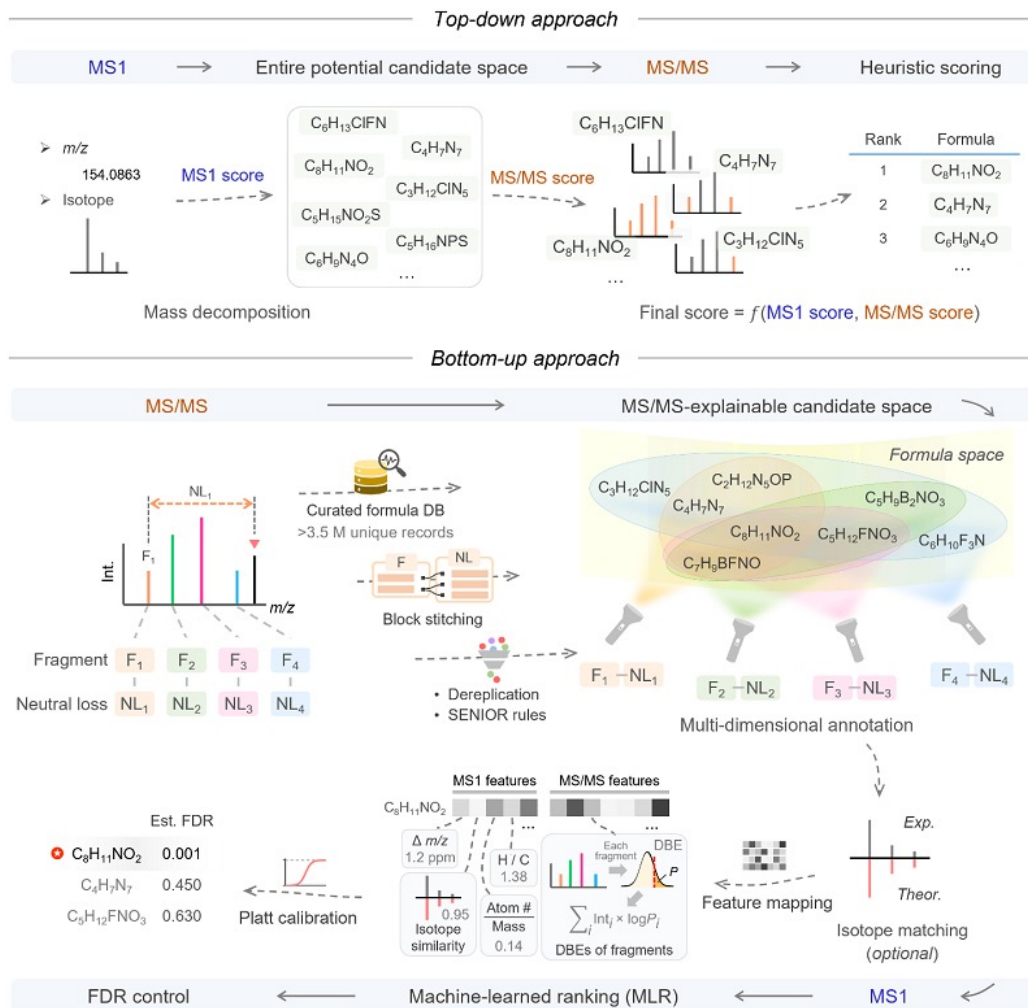
MS isotope/adduct

192.0757

Level 5: Exact mass of interest

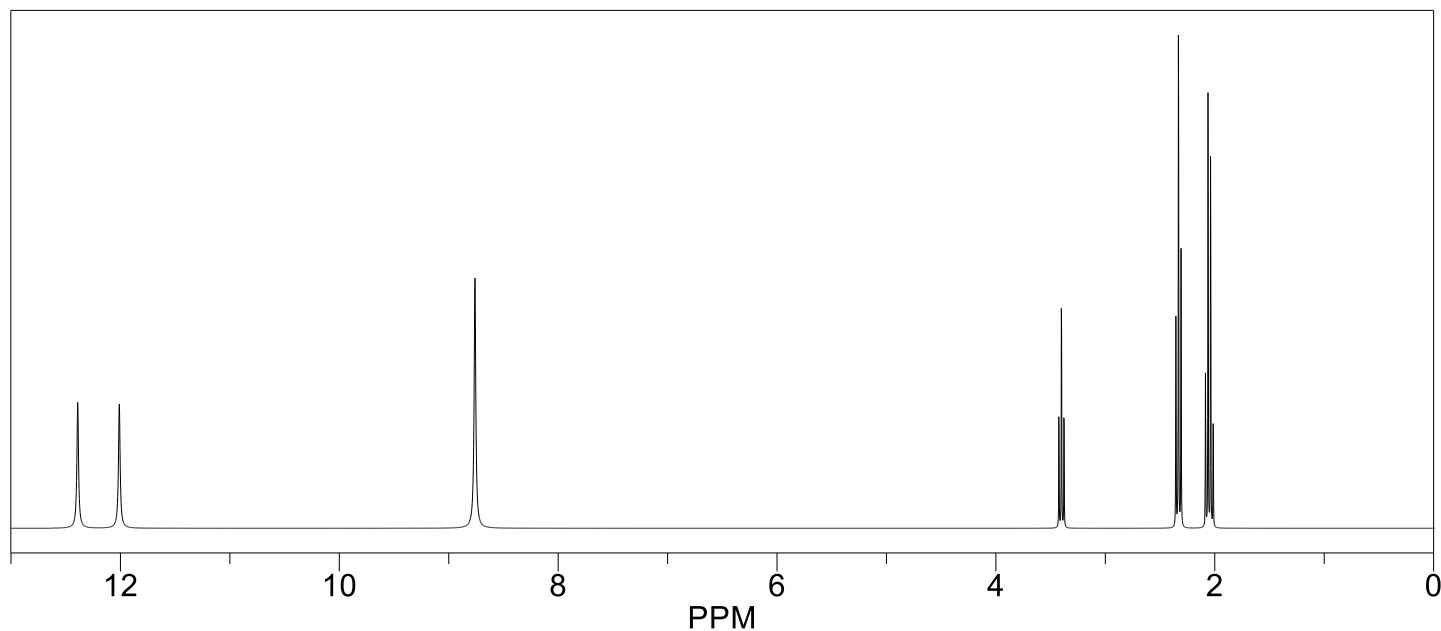
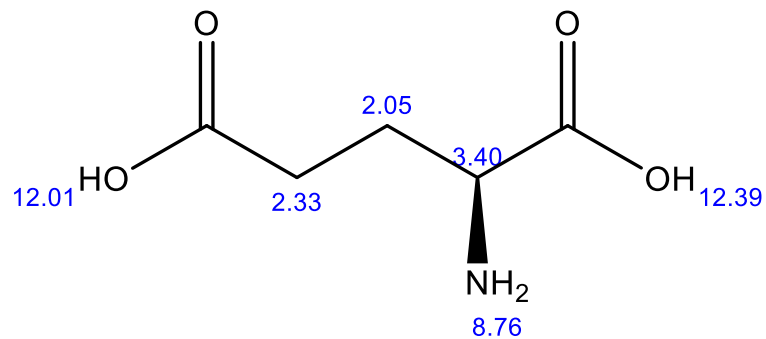
MS

BUDDY: molecular formula discovery via bottom-up MS/MS interrogation



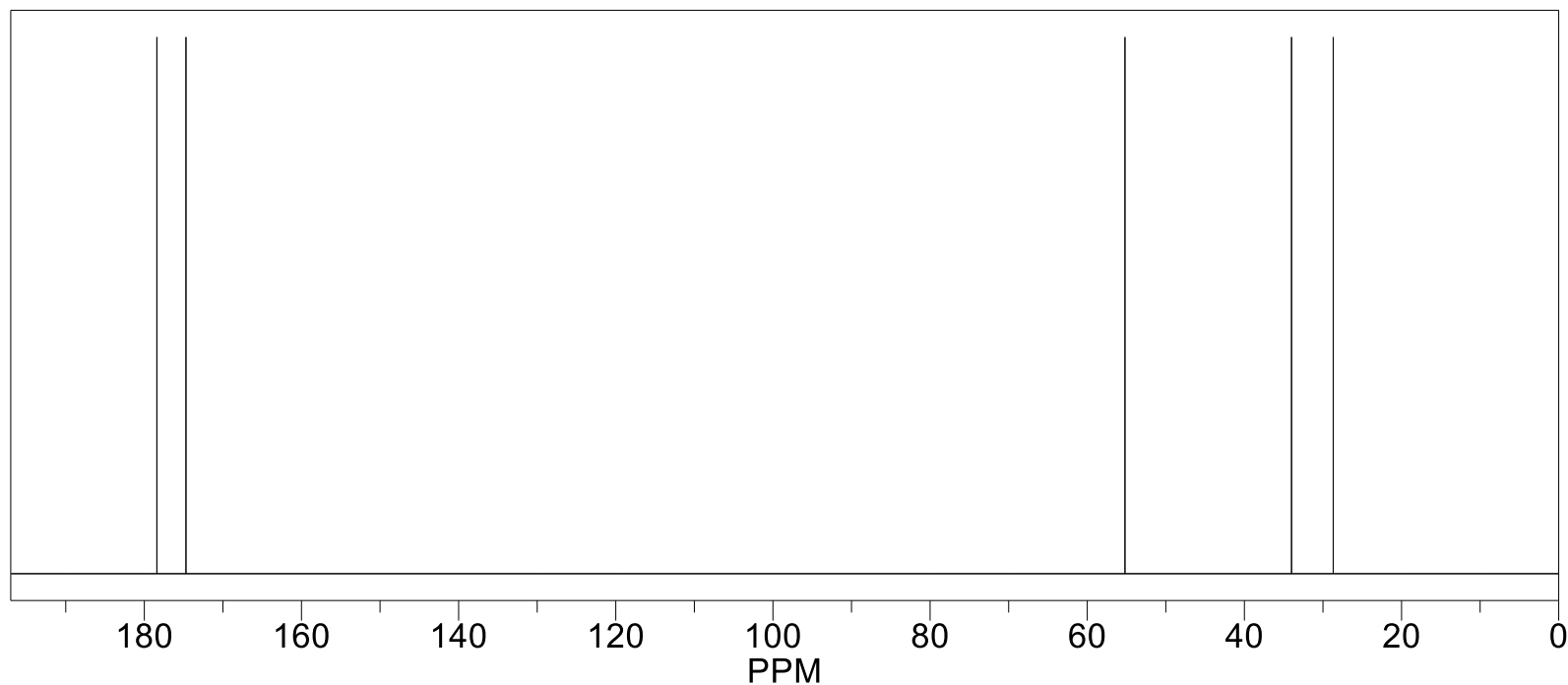
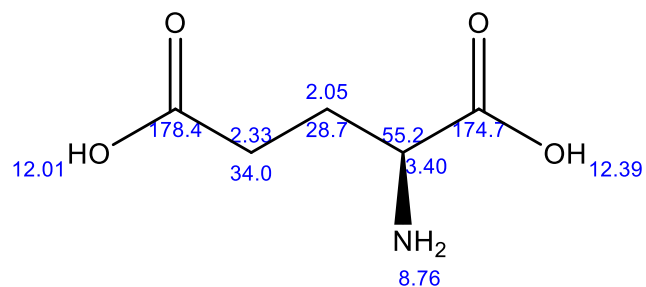
代谢产物的鉴定

■ ^1H NMR



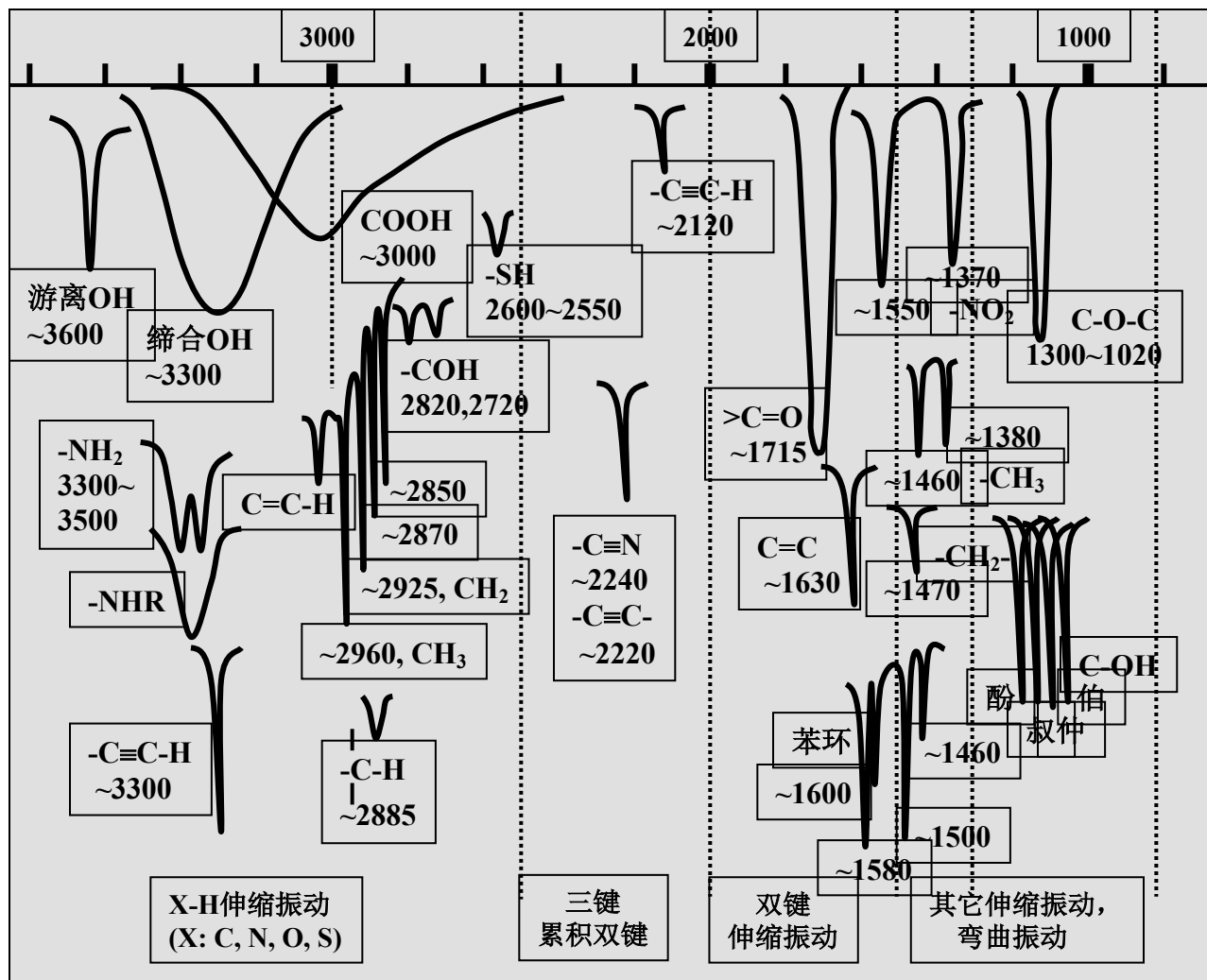
代谢产物的鉴定

■ ^{13}C NMR



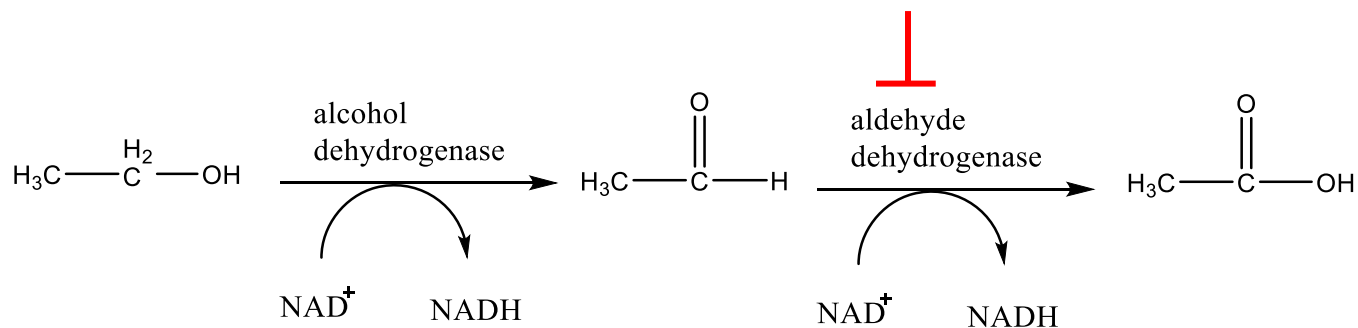
代谢产物的鉴定

■ 红外光谱



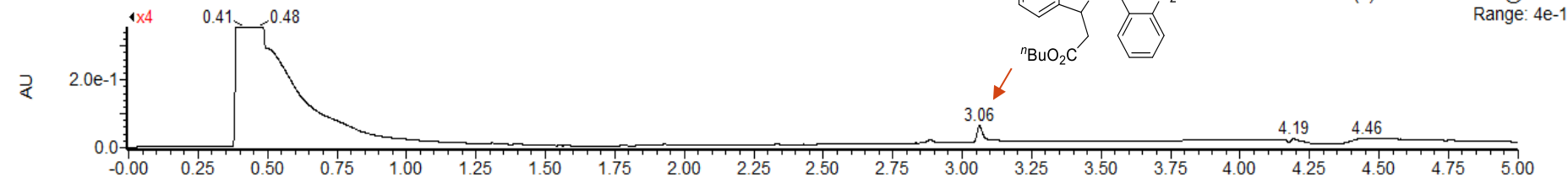
药物代谢

Cephalosporin



05-Nov-2016

LXY-9602-0min

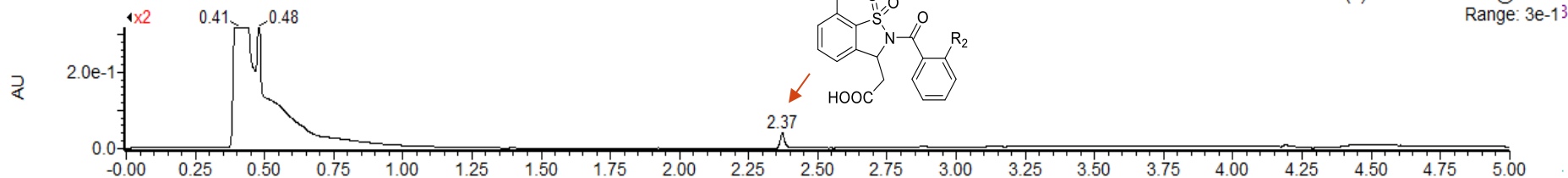


19:51:41

(1) PDA Ch1 254nm@1.2nm
Range: 4e-1

15-Nov-2016

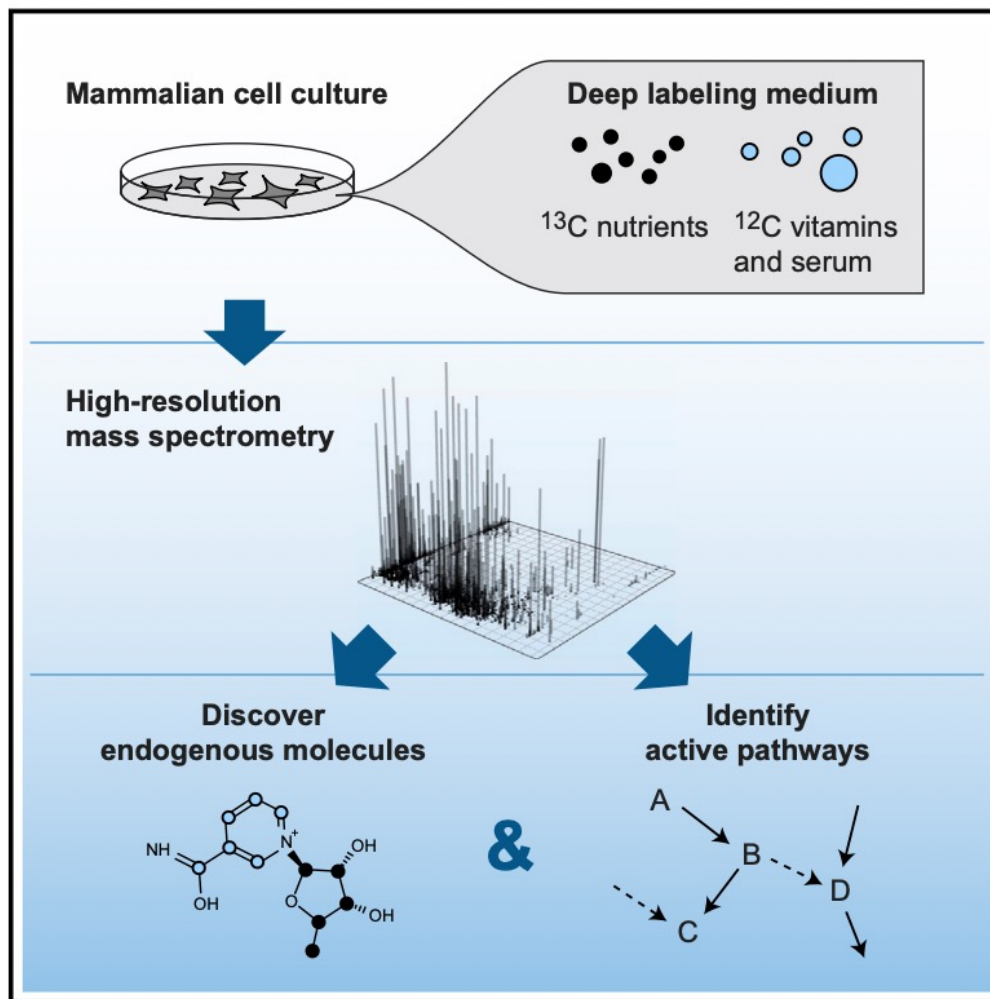
XY-9602-5min



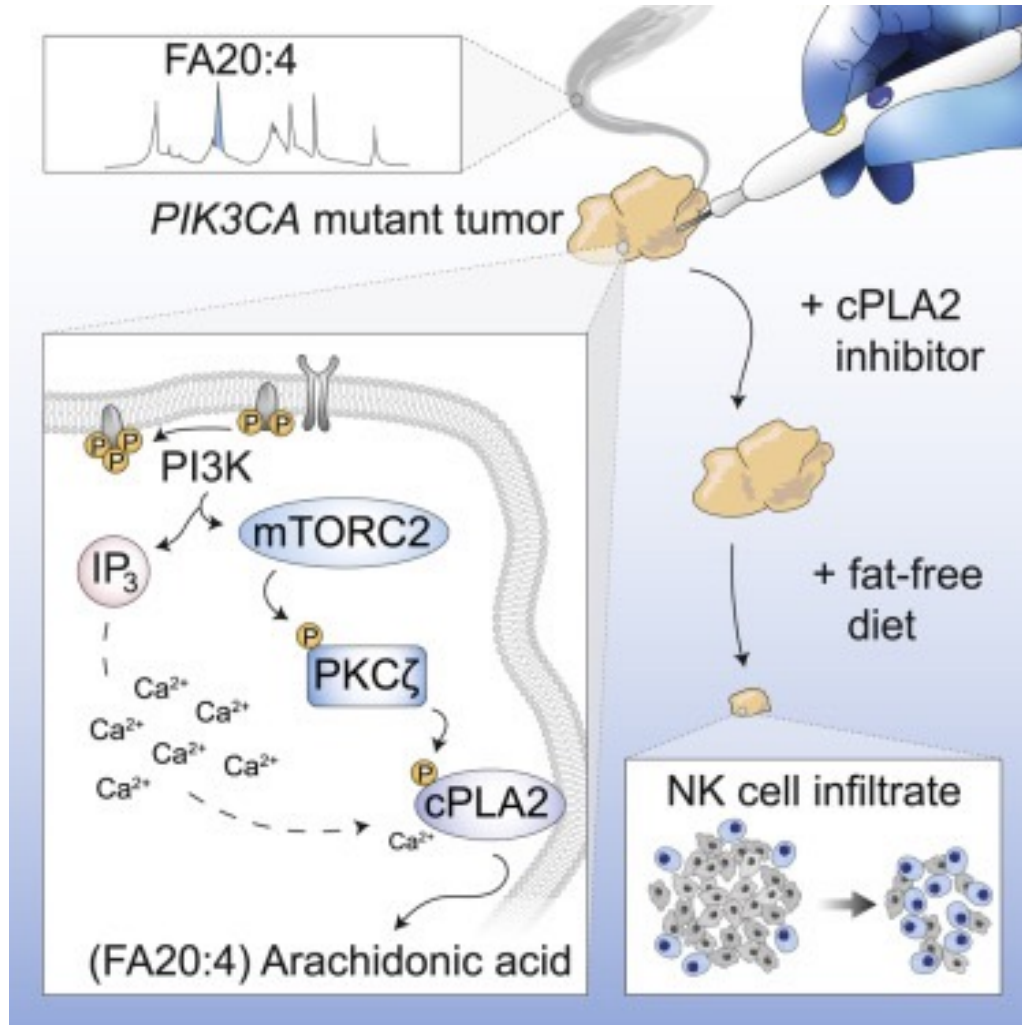
19:57:37

(1) PDA Ch1 254nm@1.2nm
Range: 3e-13

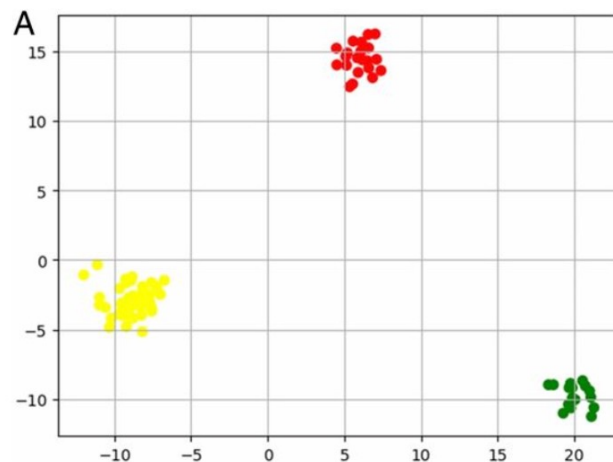
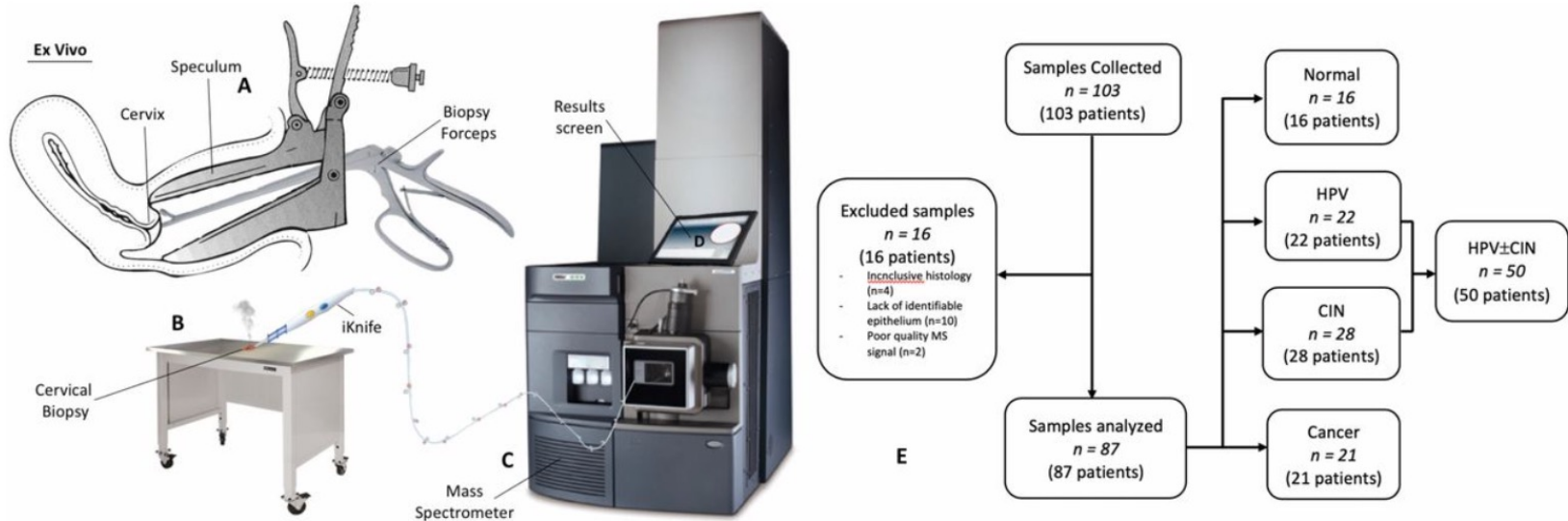
Profiling the Metabolism of Human Cells by Deep ^{13}C Labeling



Metabolic Fingerprinting Links Oncogenic *PIK3CA* with Enhanced Arachidonic Acid-Derived Eicosanoids



Intelligent knife (iKnife) diagnosis of cervical disease



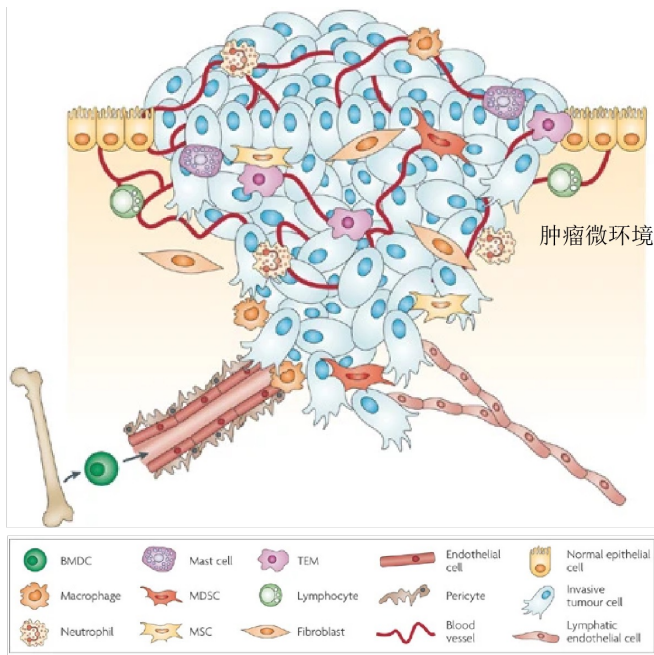
B

Proportion (%) of classification when correlating iKnife diagnosis to gold standard histology				
Predicted iKnife diagnosis	Histopathological Diagnosis			Cross Validation
	Normal (n=16)	HPV ± CIN (n=50)	Cancer (n=21)	
Normal	100% (16/16)	0% (0/50)	0% (0/21)	100%
HPV±CIN	0% (0/16)	100% (50/50)	0% (0/21)	
Cancer	0% (0/16)	0% (0/50)	100% (21/21)	

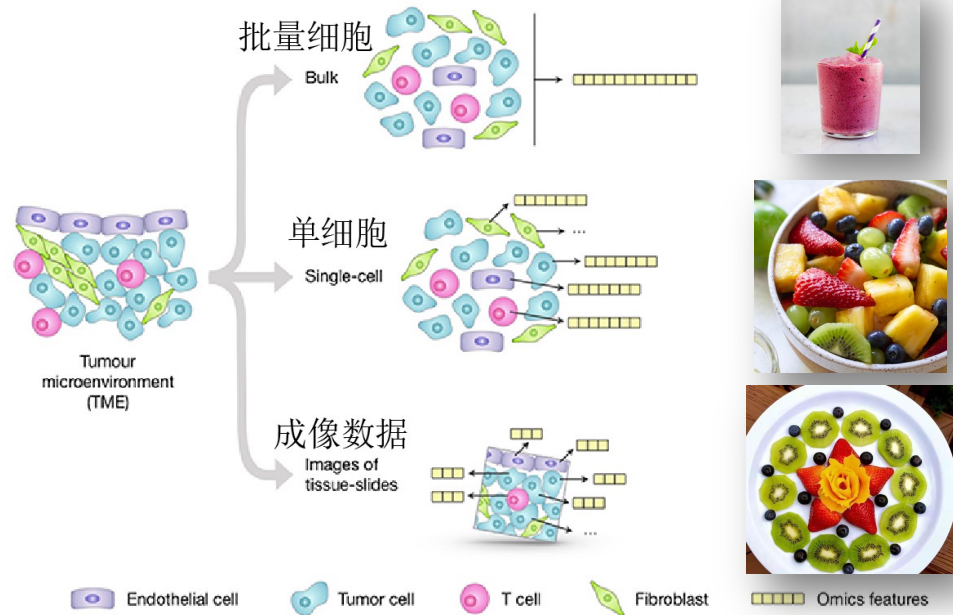
- Normal Cervical Tissue
- HPV ± CIN
- Cancerous Cervical Tissue

组织时空信息对于生物医学研究具有重要意义

肿瘤微环境 (Tumor microenvironment, TME)

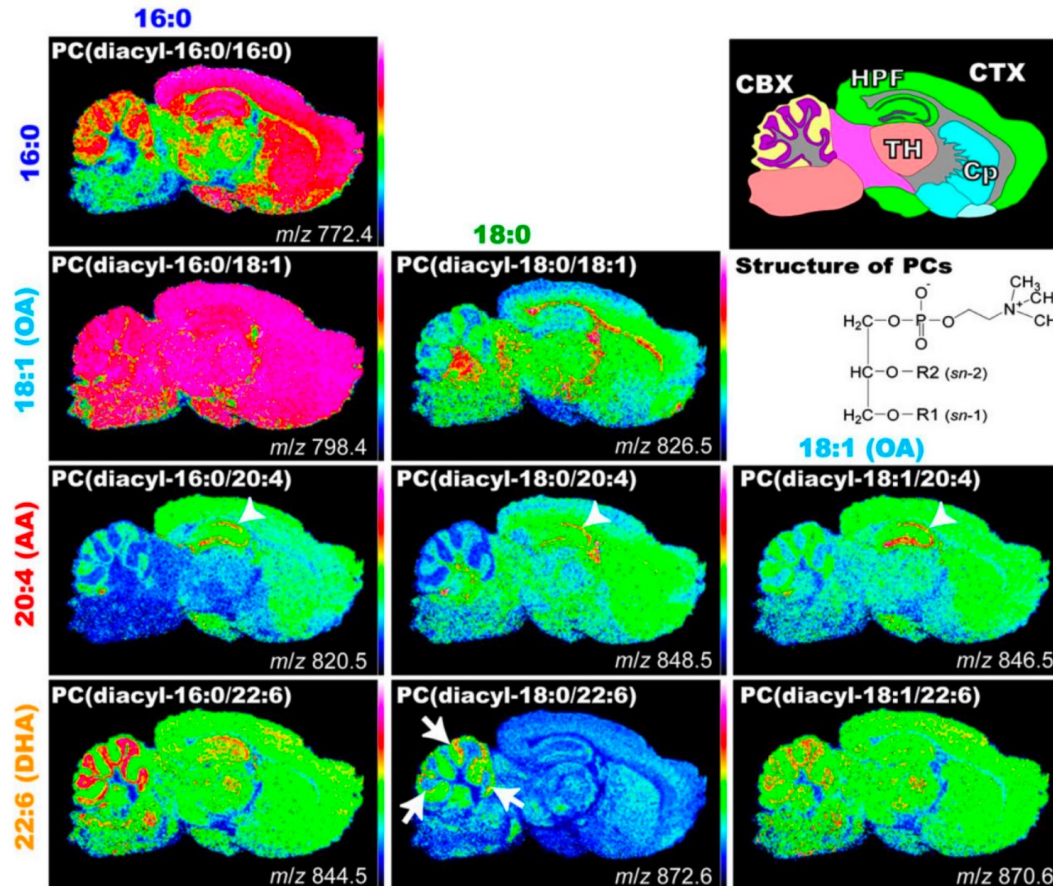


单细胞组学技术发展



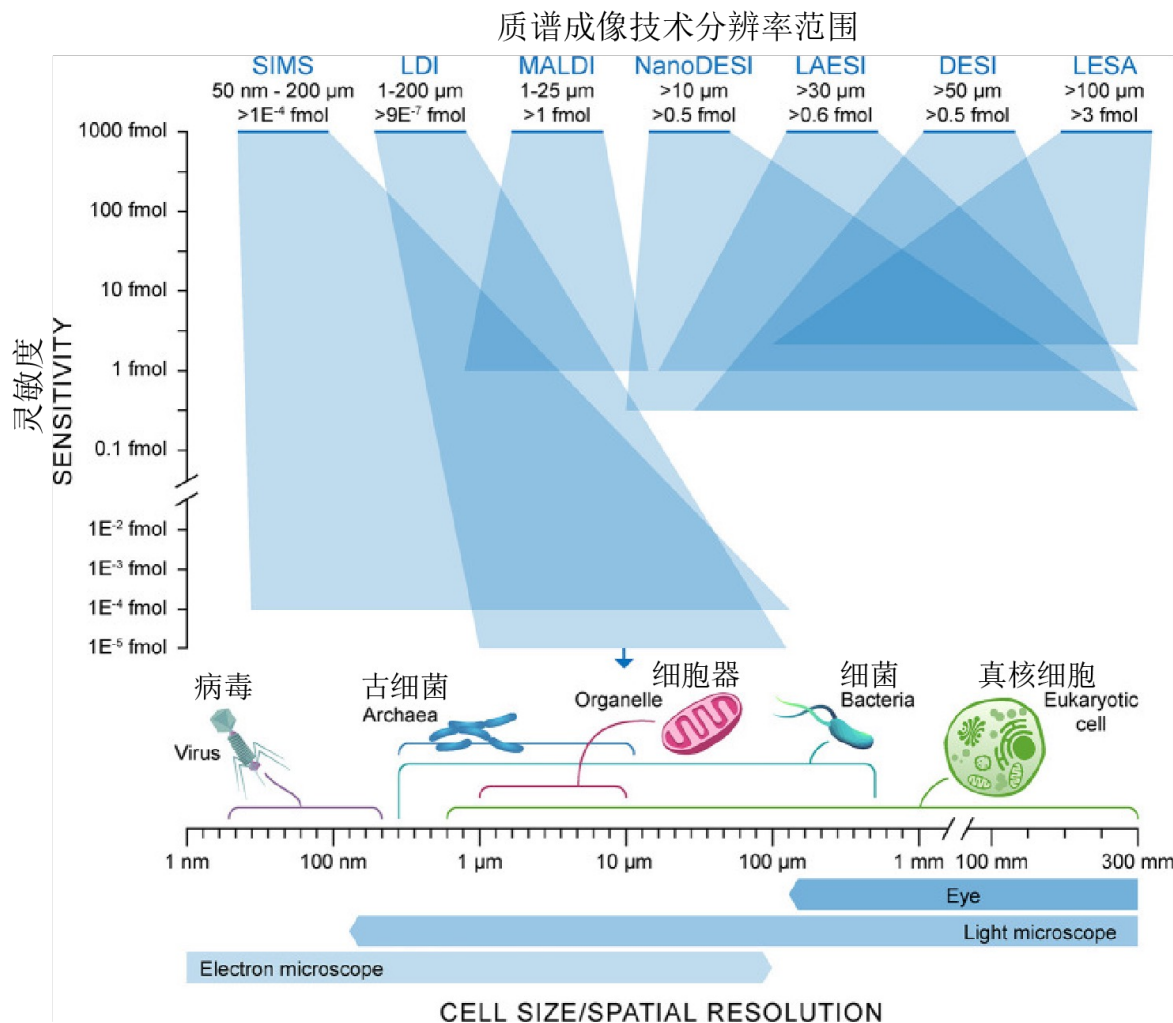
Finotello, F. & Eduati, F. *Frontiers in Oncology* **8**, (2018).
Joyce, J. A. & Pollard, J. W. *Nature Reviews Cancer* **9**, 239-252 (2009).

组织微环境与空间质谱成像技术



□ MALDI-MS 图像展示了不饱和脂肪酸的磷脂酰胆碱在脑内的独特空间分布

质谱成像技术分辨率比较

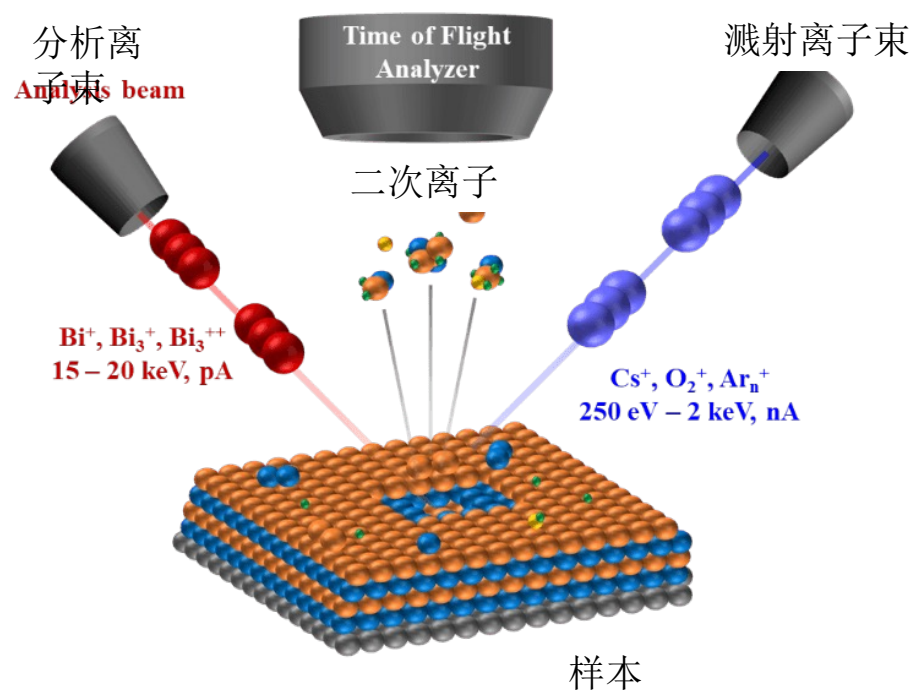


二次离子质谱技术

PHI *nano*TOF II TOF-SIMS



二次离子质谱 (SIMS) 技术原理



空间单细胞代谢组技术

nature|methods

ARTICLES

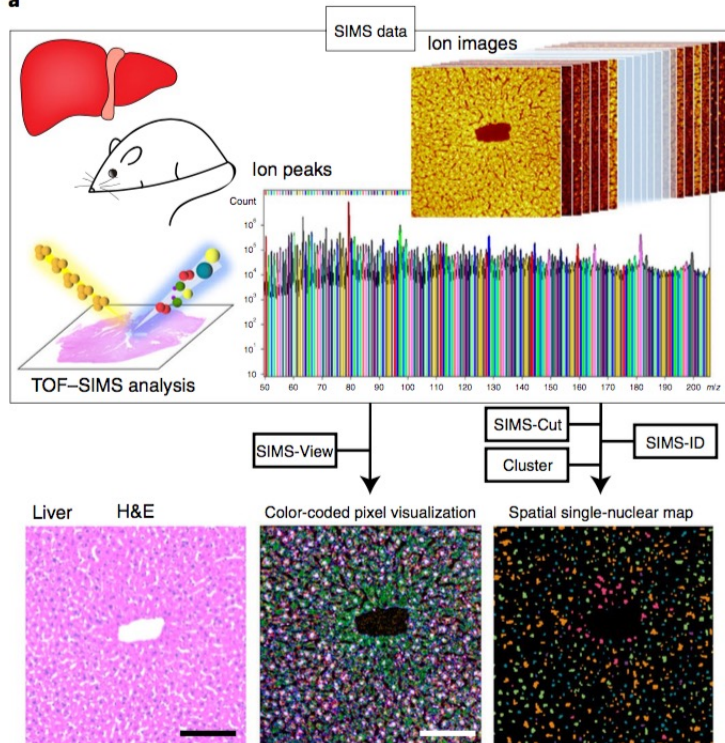
<https://doi.org/10.1038/s41592-021-01276-3>

Check for updates

SEAM is a spatial single nuclear metabolomics method for dissecting tissue microenvironment

Zhiyuan Yuan^{1,10}, Qiming Zhou^{2,10}, Lesi Cai^{3,10}, Lin Pan⁴, Weiliang Sun⁴, Shiwei Qumu⁵, Si Yu⁶, Jiaxin Feng³, Hansen Zhao³, Yongchang Zheng⁶, Minglei Shi⁷, Shao Li¹, Yang Chen^{1,9,11,12}, Xinrong Zhang^{3,11,12} and Michael Q. Zhang^{1,7,8,11,12}

a



news & views

Check for updates

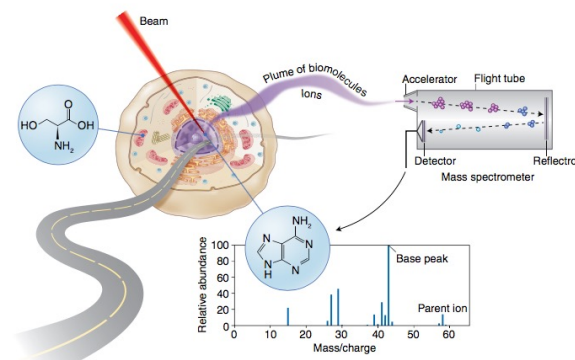
MASS SPECTROMETRY

Mass spectrometry comes of age for subcellular organelles

Technological innovations in optical object recognition and high-throughput ultrasensitive mass spectrometry are enabling subcellular metabolomics and peptidomics, providing unprecedented opportunities to study small-molecule mediators of cellular function with important implications in health and disease.

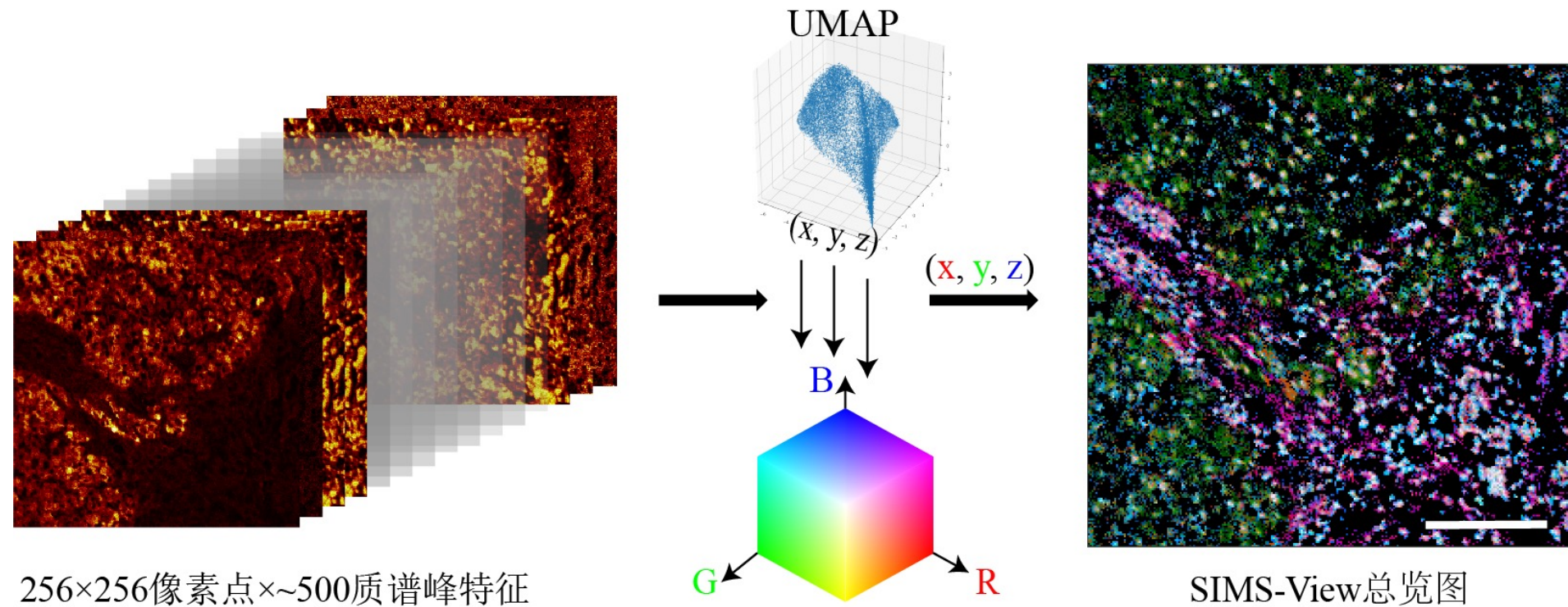
Peter Nemes

Increasing evidence from single-cell studies has revealed the importance of characterizing molecular heterogeneity not only between cells, but also within cells, in order to fully understand cellular complexity and fitness. Even slight perturbations to the (sub)cellular expression of genes and their associated networks with proteins, peptides and metabolites can trigger profound changes in cellular homeostasis, activity and function or capacity for differentiation. High-resolution imaging and deep sequencing allow quasi-routine screening of genes and transcripts across large populations of cells and organelles, driving new discoveries¹. However, for functionally important proteins, peptides and especially for metabolites, limits to the sensitivity of mass spectrometry (MS) — the technology of

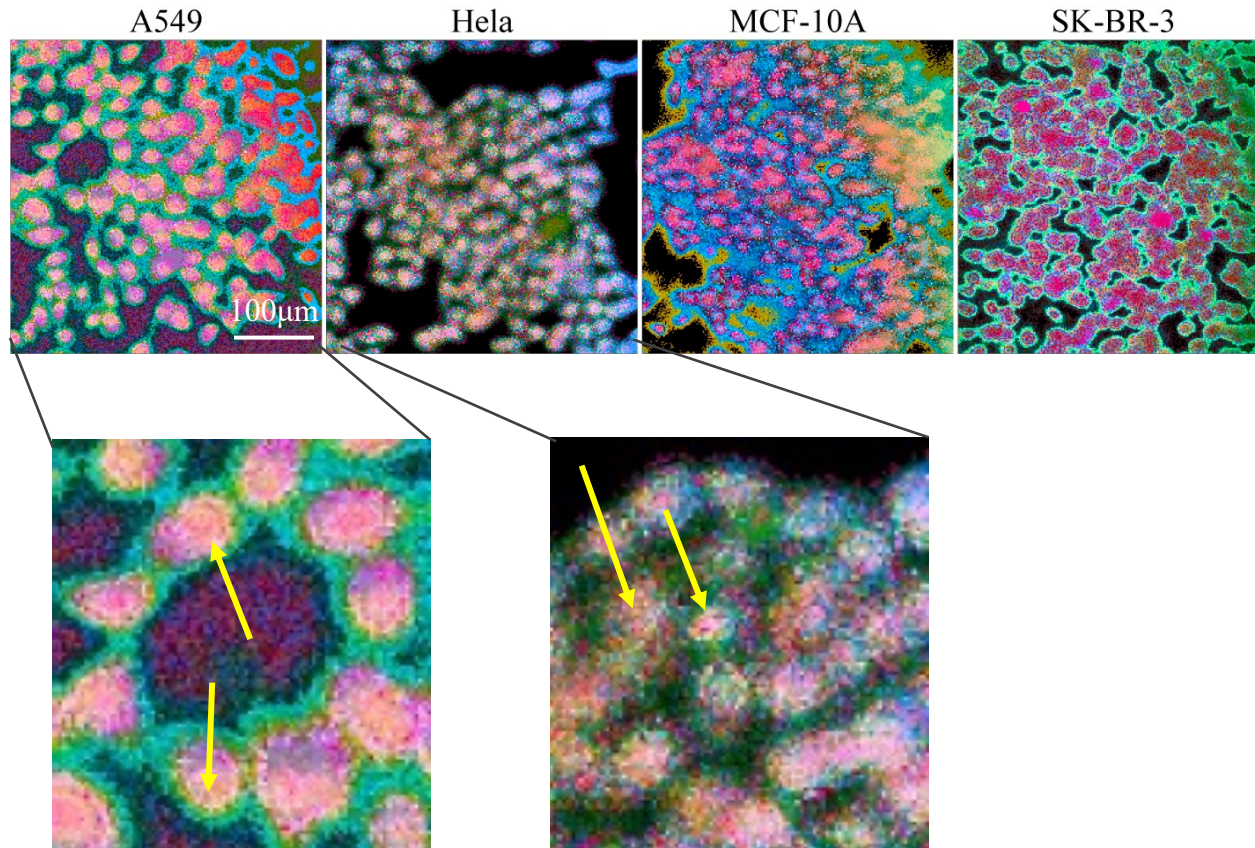


- ❑ Nature Methods发表同期专评文章，称本方法推动了质谱成像技术进入到亚细胞器的时代。
- ❑ 入选**2021年中国生物信息学十大进展**，评价该方法为首个组织空间单细胞核代谢组技术。

SIMS-View降维可视化空间代谢组数据

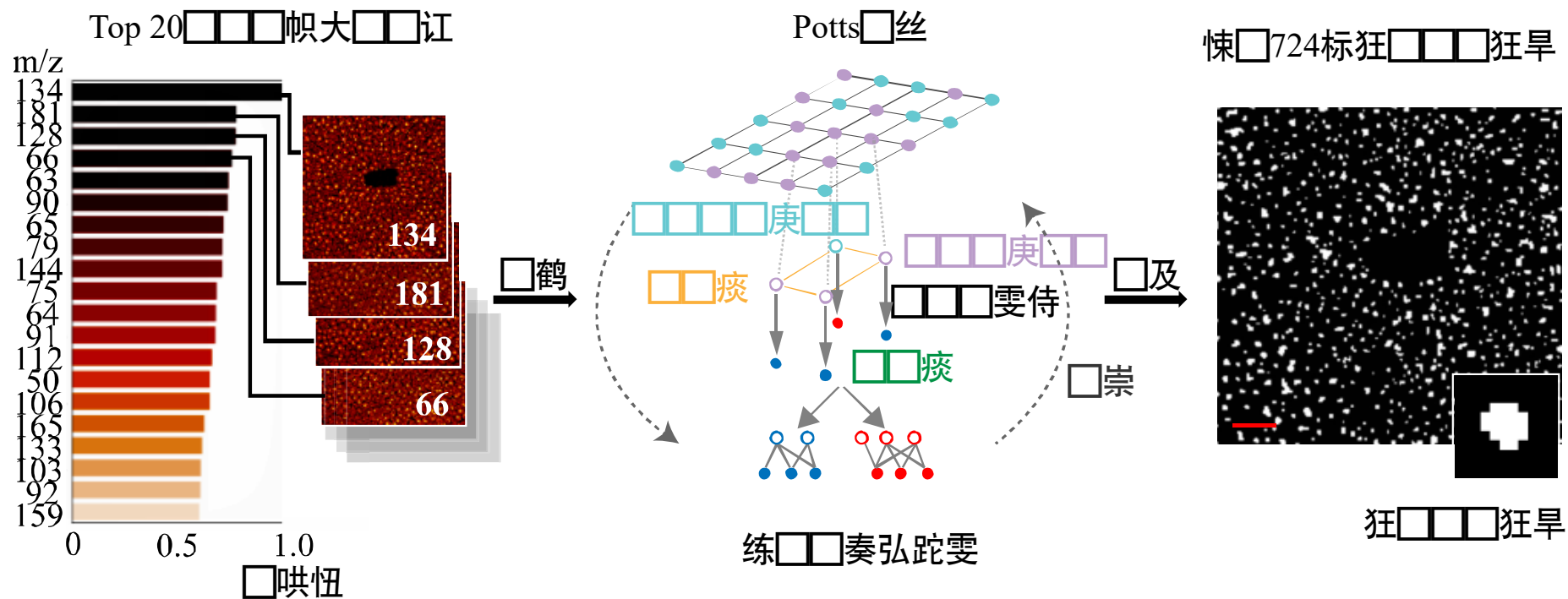


SIMS-View降维可视化空间代谢组数据



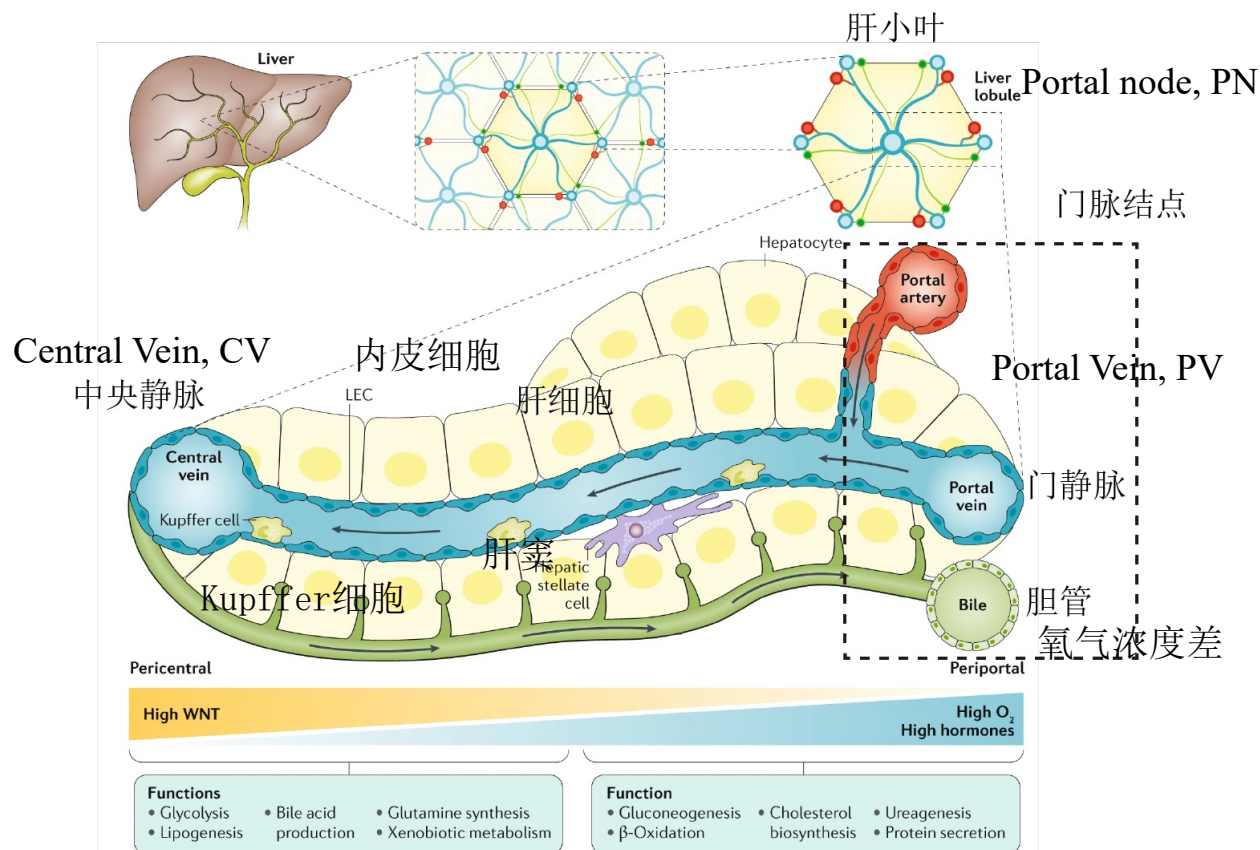
细胞核颜色相似（特征相似）

单细胞核识别算法SIMS-Cut



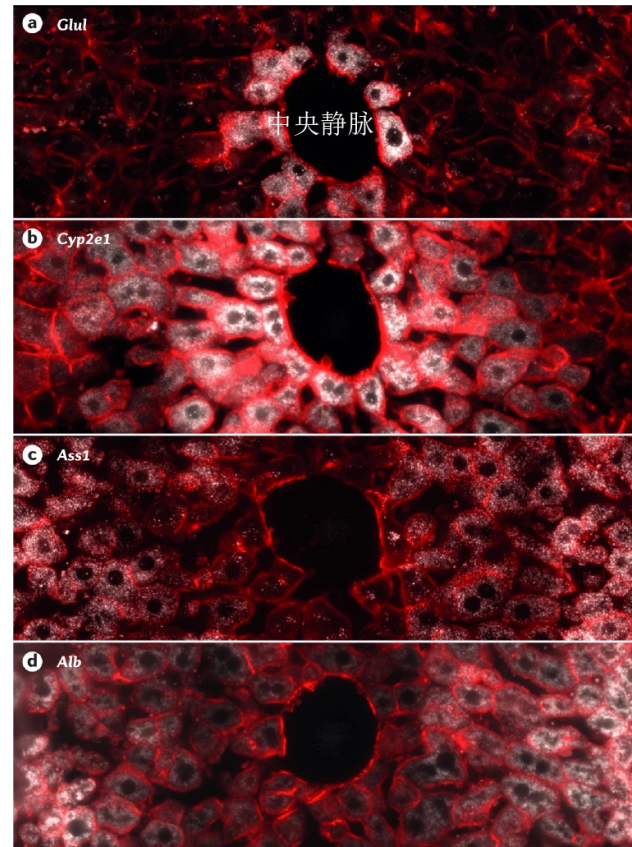
SEAM方法在小鼠组织中的测试

► 正常肝组织具有肝脏分区化 (Liver zonation) 现象

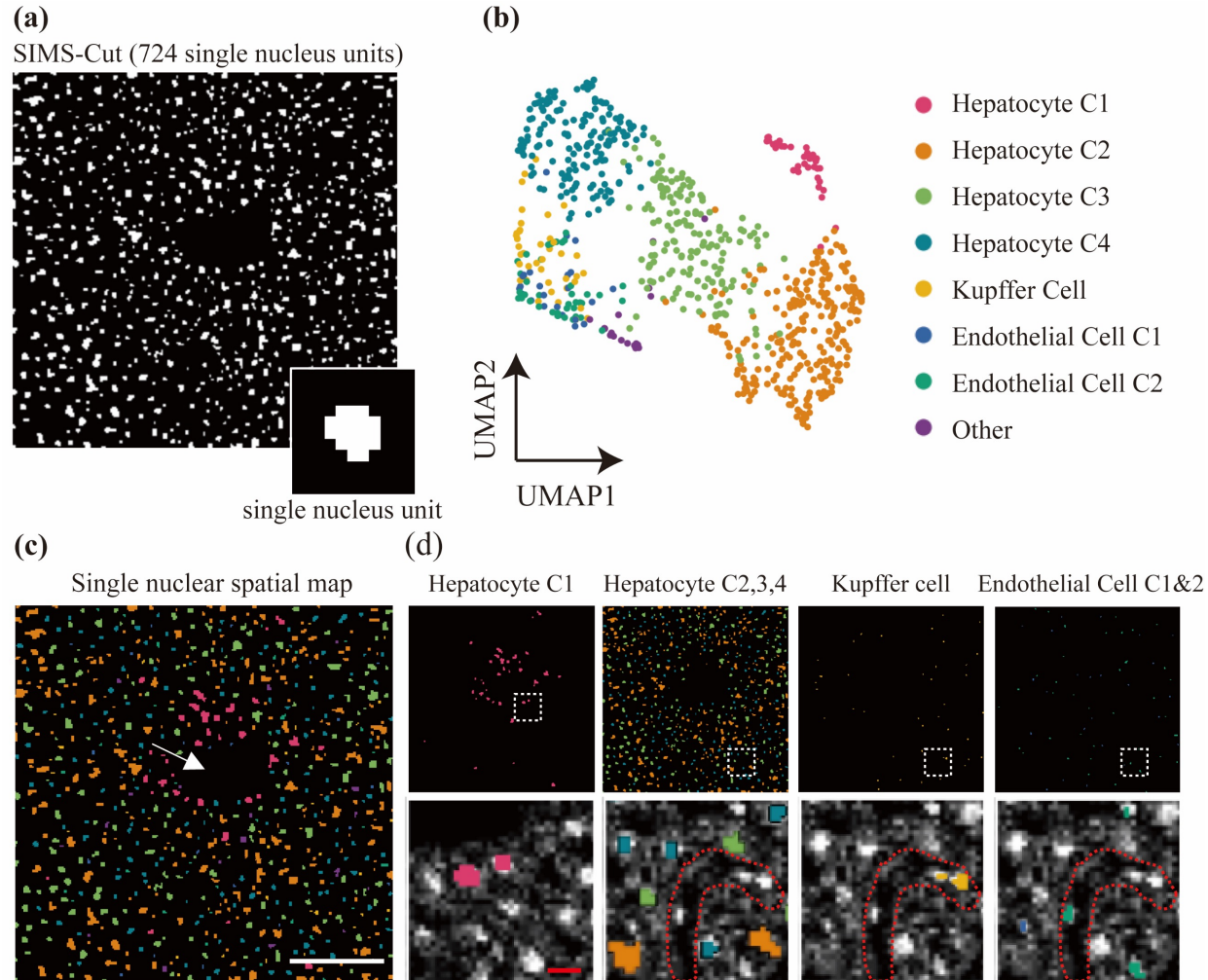


(a)

肝脏分区标志基因

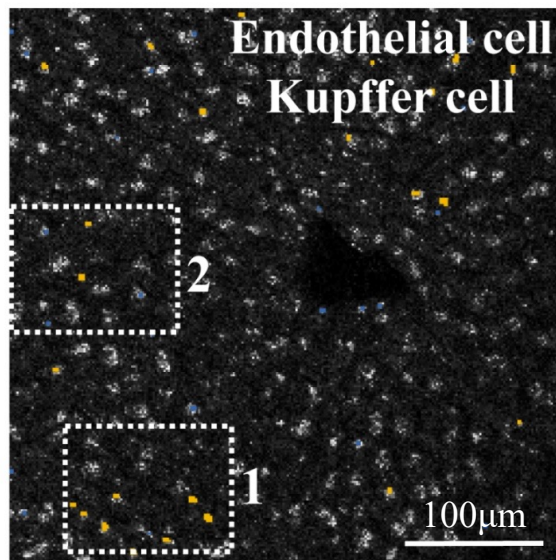


SEAM鉴定小鼠肝组织不同细胞类型

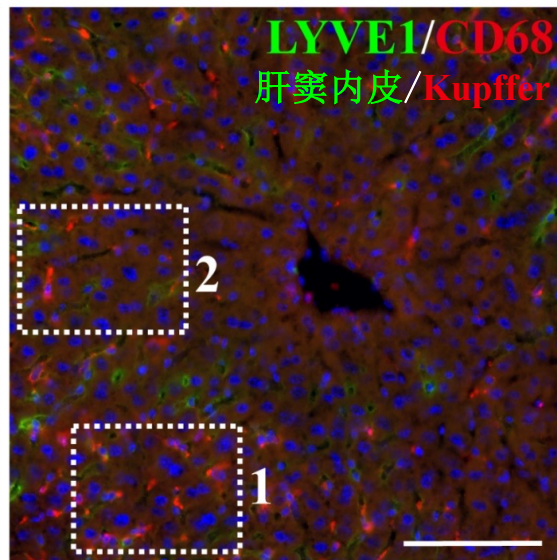


SEAM鉴定小鼠肝组织不同细胞类型

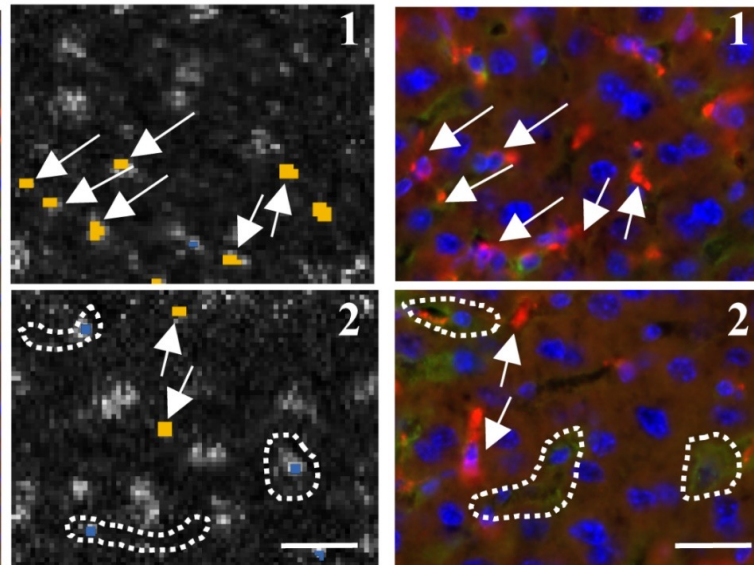
SEAM聚类结果



免疫荧光



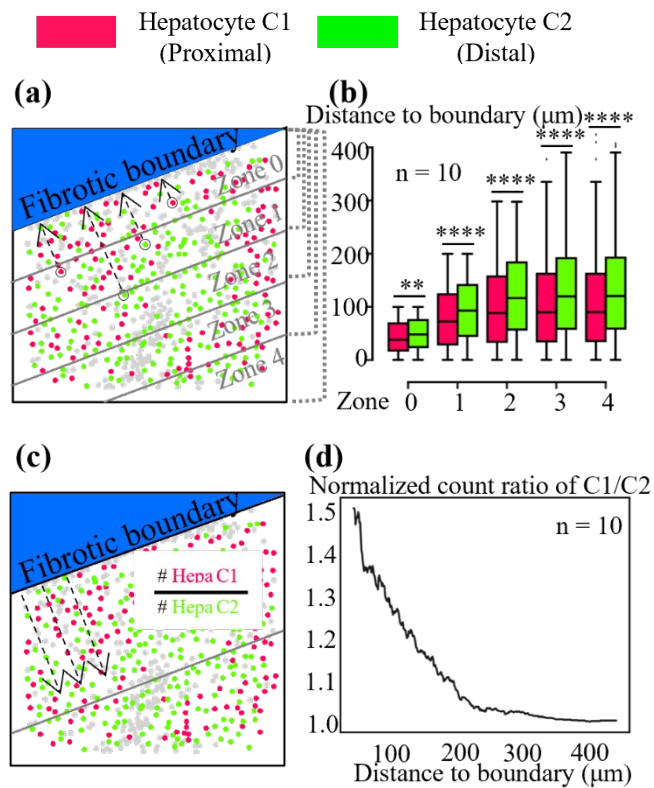
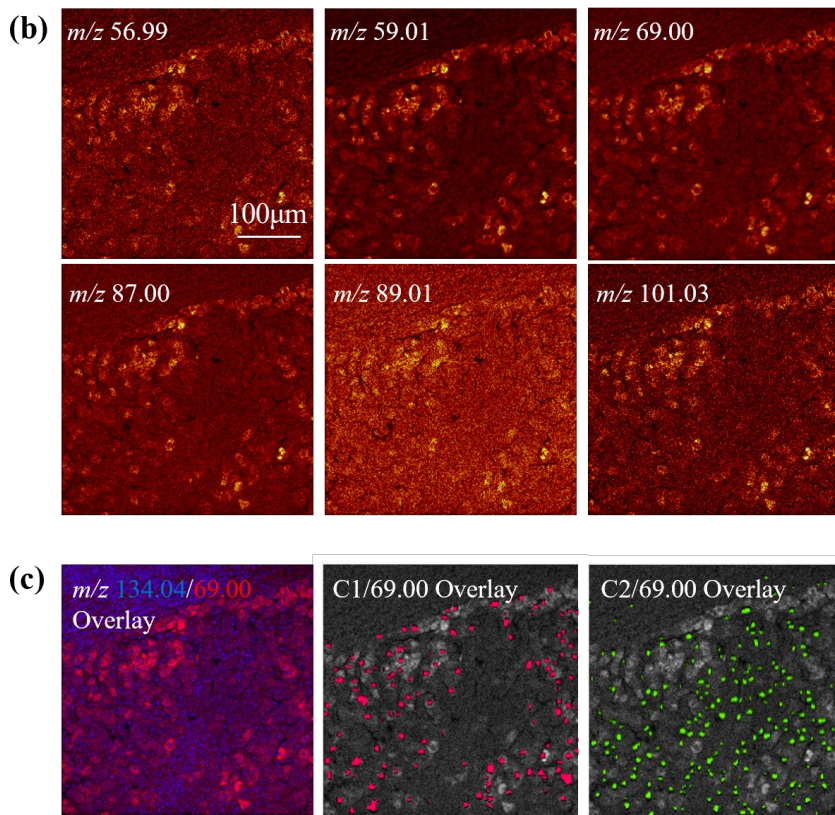
肝窦内皮细胞
Kupffer细胞



我们发现 m/z 59.96、75.96 和 76.96 与内皮细胞共定位，而 m/z 134.04、180.89 和 90.98 在 Kupffer 细胞中富集

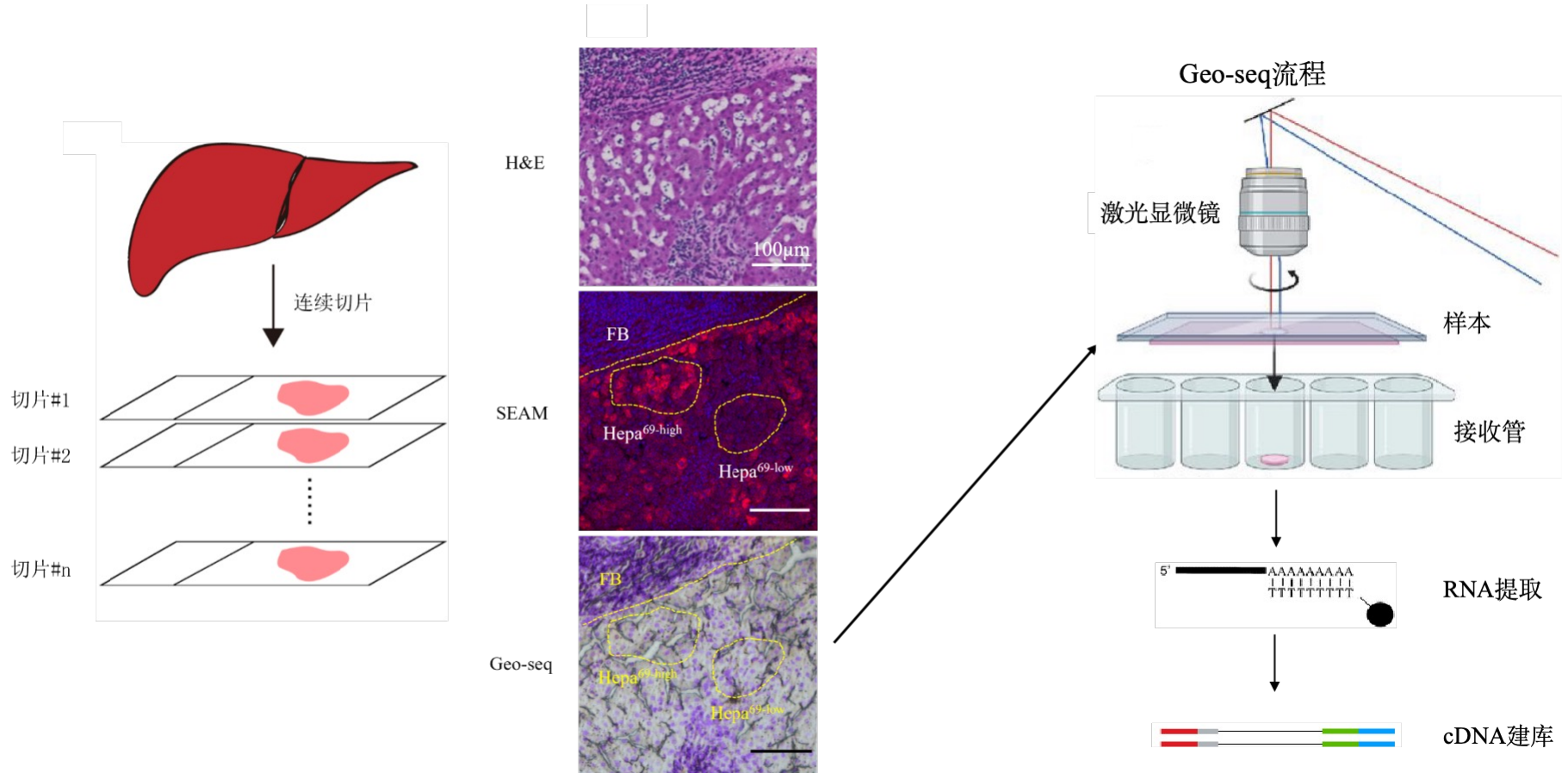
SEAM可以通过代谢特征所发现小鼠肝组织中不同的细胞类型

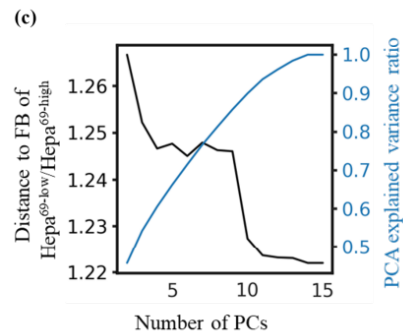
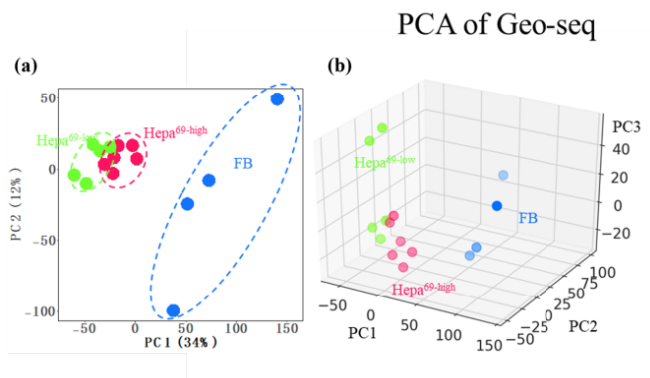
SEAM方法在人纤维化肝组织中的应用



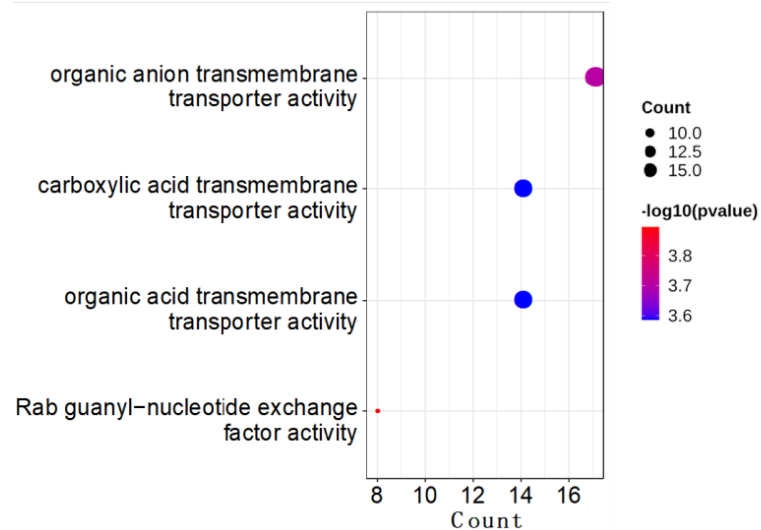
肝细胞亚群的空间分布差异和纤维化微环境之间具有关联性

联用空间转录组验证肝细胞亚群存在代谢相关基因差异表达

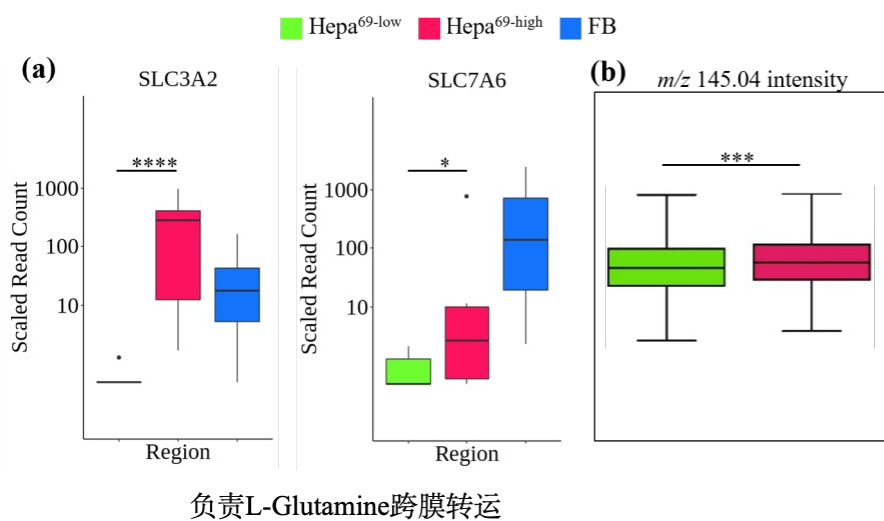




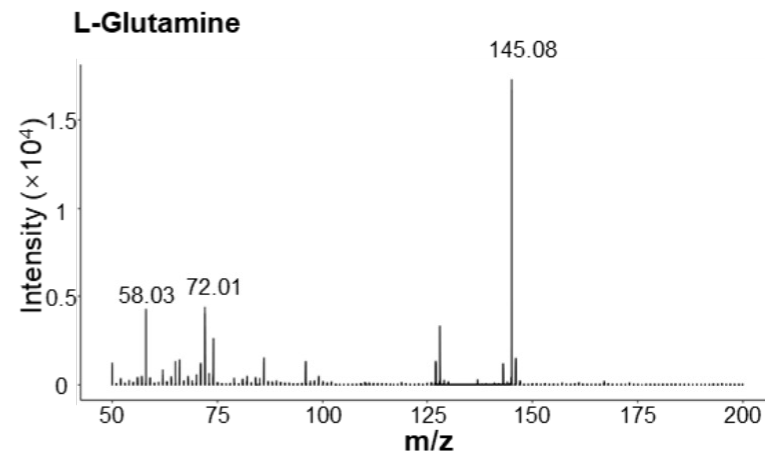
Hepa^{69-high} vs Hepa^{69-low} 差异表达基因GO通路富集



空间转录组验证肝细胞亚群有代谢相关基因差异表达



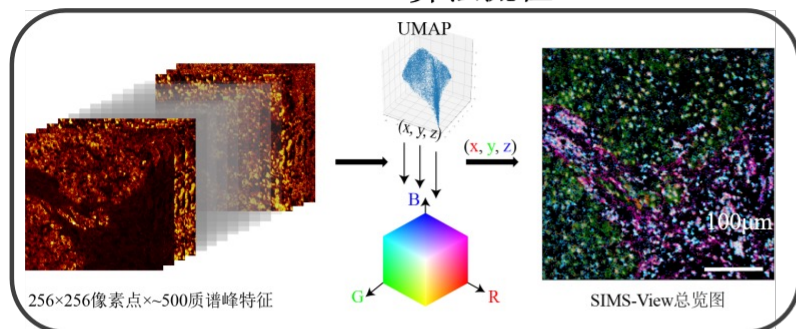
标准品质谱峰



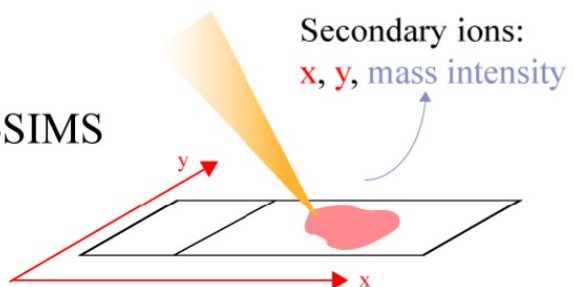
SEAM算法在其他组学方法上的扩展应用

➤ SIMS-View可视化空间组学数据异质性

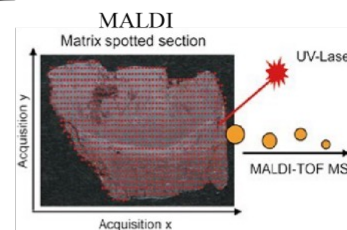
SIMS-View算法流程



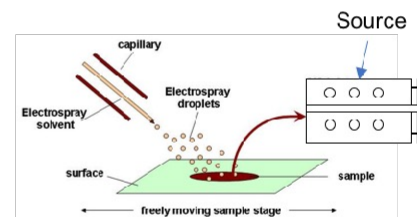
TOF-SIMS



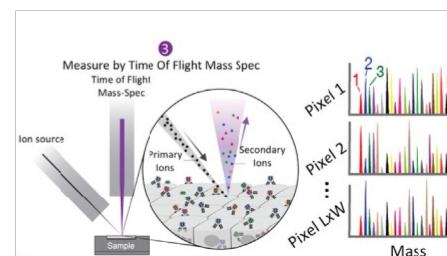
Interchangeable
to any feature



MALDI



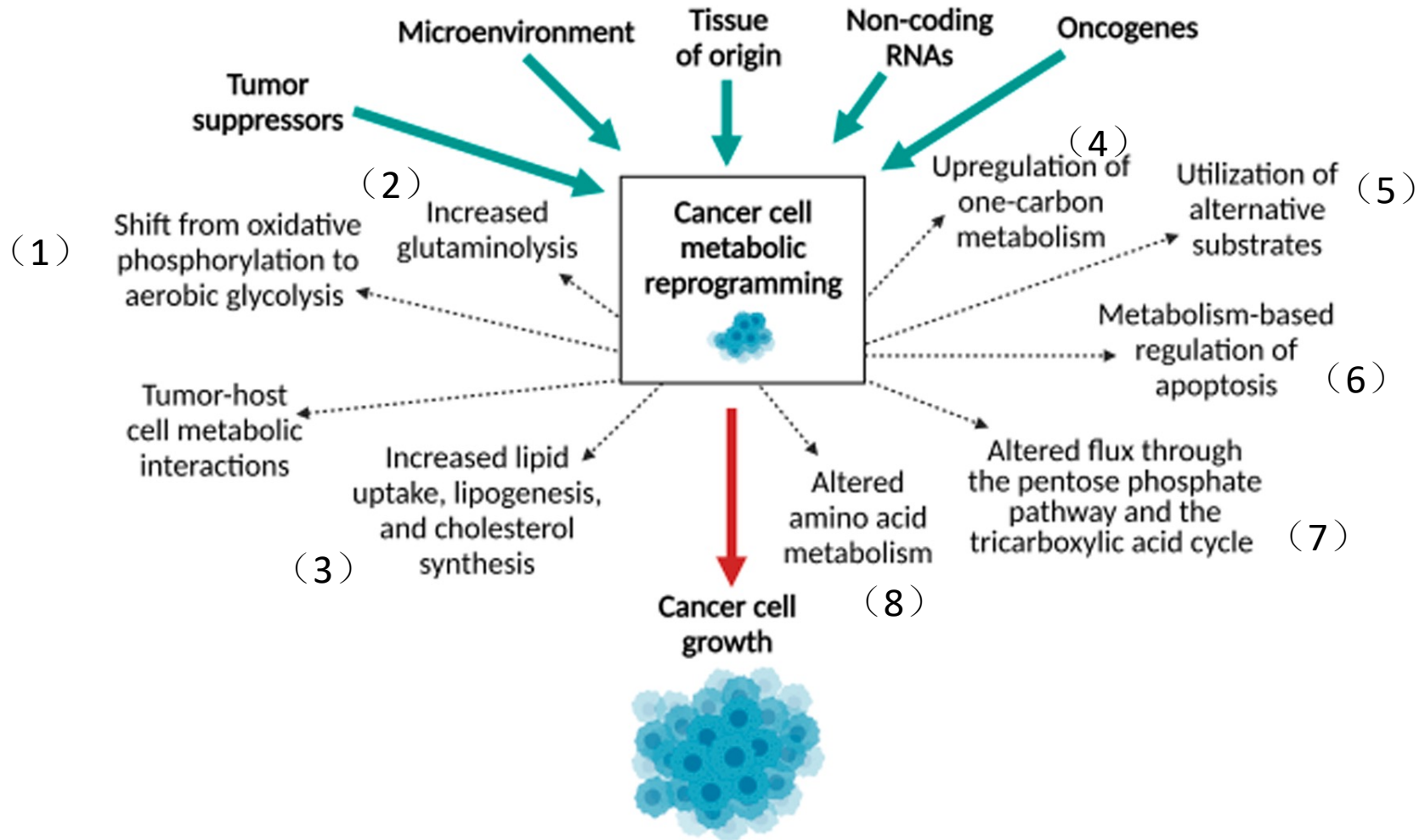
DESI



MIBI

Metabolic Reprogramming in Cancer Cells

Supporting Enhanced Growth and Proliferation



Thanks for attention!